# Exploring the Implementation of Ecotourism and Environmental Sustainability Policies: A Case Study of the Hikkaduwa Coastal Area, Sri Lanka

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# ABSTRACT

This study examines the overview of how ecotourism and environmental sustainability policies are being implemented in Sri Lanka's coastal region in Hikkaduwa. Although a big coastal tourism hub, Hikkaduwa's economy relies on beaches, coral reefs; marine tourist related activities and local livelihoods are under growing pressure caused by unplanned development, pollution, reef degradation, weak local environmental management and visitors behaviour. This study uses a qualitative case study design to understand the implementation of formal tourism and environmental policies at destination level. Data were collected through analysis of relevant policy documents, government reports and tourism planning materials, together with semi-structured interviews with four key stakeholders representing tourism regulation, coastal conservation, local government and local boat operations. It has been found that there are multiple policy and management instruments related to coastal activities in Hikkaduwa such as coastal permits, setback measures, environmental assessment, protected area management, sustainable tourism and pollution-control measures. However, implementation is inconsistent. The major problems are low monitoring, lack of institutional coordination, lack of staffing and funding, non-uniform compliance of the tourism business, poverty, reef damage, and problems with tourist behavior. The study thus concludes that it is not a lack of policies, but rather the implementation gap, and that this gap is one of the primary concerns. In order to be effective, the implementation needs greater inter-agency coordination, regular monitoring, increased participation of stakeholders, greater increased accessibility of tourism operators in the local community and increased awareness and access to practical support provided to local tourism operators. The study adds to the knowledge on the development of destination-level sustainability governance in the coastal regions of Sri Lanka and emphasizes the need for integration of conservation, tourism developments and livelihoods that balance these issues well.



**Keywords**: Ecotourism, Environmental Sustainability, Policy Implementation, Hikkaduwa, Coastal Tourism, Sri Lanka, Sustainable Tourism Governance.

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# CHAPTER ONE

## 1.1 Introduction

Tourism has emerged among the most significant sectors of the economy in the world and its development has integrated environmental sustainability as one of the major policy issues. The World Travel and Tourism Council states that the industry generated US 10.9 trillion to the world GDP and provided 357 million employment in 2024, about 10% of the global economy and one in ten employment worldwide. Simultaneously, UN Tourism indicated that by 2024, the number of international tourists returned to its normal level before the pandemic, which proves the pace and magnitude of the world tourism growth (World Travel and Tourism Council, 2024).

As United Nations reports, sustainable tourism now consist of the 2030 Agenda directly connected with SDGs 8, 12 and 14 since tourism is both an influencer of employment and a factor on the use of resources and ecosystem protection (United Nations, 2026b). However, the industry also places a significant burden on the environment. In its business-as-usual scenario, UNEP estimates that tourism may raise energy consumption by 154%, greenhouse gas emissions by 131%, water consumption by 152%, and solid waste by 251% in 2050 (UNEP, 2026b).

This concern is strengthened by a recent scholarship. Gössling *et al.* (2023) believes that until the increased mitigation, tourism may deplete the available 1.5C carbon cap globally by 40% of the total, whilst Miller and Torres-Delgado (2023) debate that the evidence based destination management cannot be achieved without sound sustainability indicators. Practical reactions are already apparent: Certification of Sustainable Tourism in Costa Rica has been applied in the measurement and enhancement of the conditions of businesses environmentally and socially and demonstrates how sustainability can move from policy to practice (United Nations, 2026a).

Evidence was found in Albania which has seen UNEP record that following a booming tourist development, a country that had developed 17.5% of its national area into protective zones has grown this to 21.4% in its ecotourism managing to reconcile recreation and conservation as well as local livelihoods using ecotourism like the Vjosa Wild River National Park (UNEP, 2025). These sources combined demonstrate that sustainable tourism offers the larger development framework, whereas ecotourism can be viewed as a more specific and conservation-based model of operationalising sustainability in the natural destinations.

The type of tourism which is one of the most spatially concentrated and sensitive to the environment is coastal tourism. The United Nations observes that approximately 80% of world tourism destination occurs in coastal regions and therefore this implies that beaches, reefs, mangroves and near shore waters are subjected to a disproportionately high tourism pressure. These ecosystems are economically valuable but ecologically fragile. UNEP highlights that alone the coral reefs deliver ecosystem services worth approximately US$2.7 trillion annually, and they are becoming more vulnerable to local stress factors like pollution, unsustainable tourism and poor coastal development coupled with climate change (UNEP, 2026a).

According to Jayarathne, Jayasekara and Gunawardena (2025), in the case of Hikkaduwa in particular, the regeneration of the corals has been slowed due to the ongoing human-induced pressures, whereas in the other study, it is indicated that there have been more than 1.3 million visits to marine protected areas in Sri Lanka in the last ten years, which helps to highlight the level of use. According to Perera *et al.* (2023), nature-based tourism in Sri Lanka is now in need of better planning, the ability of stakeholders to coordinate and implement its activities. The gap in the knowledge therefore is not concerned with the existence of policies itself but the effectiveness with their application of the policies within practice in Hikkaduwa.

## 1.2 Research Problem and Knowledge Gap

The following are the various 'Themes' observed in the literature that influence the perception of ecotourism and environmental sustainability. An inherent ambiguity in the concept of ecotourism is one of the themes. While ecotourism is generally associated with nature-based tourism, conservation, and engagement with communities and economic and social development, there remained no agreement on a single definition. Despite the abundance of nature- and conservation-themed definitions, Chandel and Mishra (2016) demonstrate that the concept of ecotourism is still poorly defined. Similarly, Khanra *et al.* (2021) determine several key areas such as ecological preservation, community interests, carbon impacts, and tourist behaviours in the study of ecotourism. This implies that ecotourism is a general term that does not always include a balance between the environmental, social and economic aspects of ecotourism.

Another major issue discussed is also the off-balance relationship between tourism development and environmental sustainability. Coral reefs, beaches and nearshore water require protection for the support of coastal visitor economy but also can be vulnerable to the impacts of tourism activities, including pollution, over-tourism, inadequate waste management and unplanned development. But Gössling and Scott (2018), make the case that sustainability problems stem from the fact that while reducing impacts is difficult, growth can be difficult as well. Higgins-Desbiolles (2018) raises doubts about the effectiveness of growth-based tourism models in promoting sustainability, as well. This is important for coastal destinations since environmental quality is central to their tourism product, and it's the most vulnerable to tourism.

The governance and policy implementation is another integral part of the problem of research. The enforcement, institutional co-ordination and participation of stakeholders are vital for the effectiveness of policies related to tourism and the environment. Becken *et al.* (2020) propose that when it comes to environmental policy in tourism, implementation is not always well-defined and there are not always measurable targets. Bichler (2021) is also pointing out that often the scope of local involvement in tourism governance is limited, despite the prevalence of participatory approaches. So the core problem is not just a lack of policy, but between the policy intention and implementation.

This problem can be observed in the tourism management process in the coastal areas of Sri Lanka. Sri Lanka has potential for ecotourism due to its biodiversity but outcomes are limited by habitat degradation, over use, pollution and governance (De Zoysa, 2022). Existing studies emanate primarily on environmental degradation and behaviour of the visitors/tourism pressure in Hikkaduwa. Jayaratne *et al.* (2023) find that the condition of the coral reef affects tourists' access to preferred outcomes and willingness to pay for this access, and ongoing human activity also slows coral recovery. Unplanned tourism development and coastal infrastructure are the two major factors that cause the degradation of the environment in Hikkaduwa, (Nisansala, Jayasinghe and Ranwella, 2021). Silva *et al.* (2022) also identify that "environmental degradation," or the lack thereof, is a contributor to low levels of tourist satisfaction, and is caused by unplanned tourism development. Additionally, microplastic contamination has been mentioned as a new coastal pollution problem in coastal waters (including Hikkaduwa) by Mahagamage *et al.* (2025).

Thus, one of the major research gaps is understanding the implementation of dyadic operation namely ecotourism and environment sustainability policies from a destination perspective of Hikkaduwa in practice. This study seeks to fill that gap through a study of policy documents, institutional responsibilities, experiences of actors and obstacles to policy effectiveness.

## 1.3 Aim of the Study

This study aims to examine the implementation of ecotourism and environmental sustainability policies in the Hikkaduwa coastal area of Sri Lanka. It focuses on how these policies are applied in practice, the roles of key institutions and stakeholders, and the challenges affecting effective implementation at the destination level.

## 1.4 Research Questions

To achieve the aim of this study, the following research questions are addressed:

1. What ecotourism and environmental sustainability policies are applicable to the Hikkaduwa coastal area in Sri Lanka?
2. How are these policies implemented in practice within the Hikkaduwa coastal tourism context?
3. What institutional, governance, and stakeholder-related challenges affect the implementation of these policies at the destination level?
4. What other factors influence the effectiveness of policy implementation in Hikkaduwa?

## 1.5 Overview of Methodology

This study adopts a qualitative case study approach to examine the implementation of ecotourism and environmental sustainability policies in Hikkaduwa. The research focuses on understanding how these policies are applied in practice and the challenges affecting their effectiveness. Data was collected through interviews with key stakeholders involved in tourism governance, along with the analysis of relevant policy documents. This approach enables an in-depth exploration of institutional roles, governance structures, and implementation gaps at the destination level. The findings provide insight into how policy intentions are translated into real-world outcomes.

## 1.6 Structure of the Thesis

This thesis is structured into six chapters. Chapter 1 introduces the research background, problem, aims, and research questions. Chapter 2 presents the literature review, focusing on ecotourism, environmental sustainability, policy implementation, and destination governance. Chapter 3 outlines the research methodology and methods used in the study. Chapter 4 provides the contextual background of Sri Lanka's ecotourism policies and the Hikkaduwa case study area. Chapter 5 presents the findings from the policy document analysis and stakeholder interviews. Chapter 6 discusses the findings in relation to the research questions and relevant literature, and presents the recommendations, limitations and conclusion of the study.

# CHAPTER TWO

## 2.1 Introduction

This literature review addresses both theoretical and empirical literature that would be related to ecotourism, environmental sustainability and policy implementation in the context of coastal tourism, and specifically on Sri Lanka and Hikkaduwa. It evaluates critically the conceptualisation of ecotourism, the principles and criticisms of ecotourism as well as the environmental issues surrounding the development of coastal tourism. The environmental governance, policy implementation theory and roles of stakeholders in realisation of sustainable outcomes also are analysed in the review. It is through literature synthesis on global, national, and destination related literature that this chapter has identified major knowledge gaps of how to analyse the implementation of ecotourism and environmental sustainability policies in Hikkaduwa.

## 2.2 Conceptualising Ecotourism

Conceptualisation of ecotourism is a controversial issue that is associated with its multi-disciplinarity and changing stakeholder expectations. As Chandel and Mishra (2016) find, there is no single universal and agreed definition as their article has reviewed 42 of them. Their main conclusion that nature-related traveling and conservation exist in nearly three-quarter of the definitions shows that the environmental concerns are conceptually dominant. Although this implies a certain convergence, it also demonstrates the shortcoming the relative marginalisation of socio-economic aspects relating to community empowerment and cultural integration. Even though emerging definitions progressively identify participation and local development, this move seems more of an ideal rather than the operational, thus the question arises on whether the factors are captured substantially in practice.

Similarly, in their bibliometric review of 878 studies, Khanra *et al.* (2021) determine four key research areas, including ecological preservation, community interests, carbon impacts, and tourist behaviour. Their results show that there has been a consistent shift in the environmental focus to sustainability. But this growth has only created conceptual fragmentation and not integration. The fact that various themes are presented together without a single idea indicates that ecotourism is not coherent theoretically.

Moreover, the preponderance of the research that is founded in the developed nations constrains the relevance of these findings to the developing setting, where governance and resource constraints play a paramount role in implementation. Mtapuri and Giampiccoli (2019) also state that definitional ambiguity is further enhanced by overlaps with the community-based tourism where environmental and socio-economic priorities are usually disproportionately distributed. Their community-based ecotourism (CBET) suggestion tries to balance all these dimensions trying to unite conservation with local control and equal distribution of benefits. However, its applicability is undermined by the unsupported nature of research. In general, although ecotourism has become more holistic and inclusive in paradigm, there has been central element of definitional inconsistencies and theoretical fragmentation which is indicative of a huge disparity between the conceptual development and its practical applications.

The main theories of ecotourism are normally placed in the combination of environmental conservation, the involvement of the community and economic development but their practice on the ground is not balanced. Cossengue, Brea and Tavares (2025) place ecotourism as a radical process that could produce socio-economic and environmental outcomes, especially in the creation of job and support of livelihood in the rural communities. Although such an opinion emphasizes the ecotourism value that is multidimensional, it depends on active participation of the community and good governance. The discovery by them that emotional and value-based tourist experiences are the drivers of satisfaction and loyalty adds more to the discussion that has traditionally relied on conventional sustainability measures. Even so, the gap found in between perception, intention and actual implementation identifies that these principles are not actually carried out, but made idealistic. This begs the question of how far the practical implementation of strategic planning and stakeholder coordination are implemented on the ground.

Likewise, Digun-Aweto, Fawole and Saayman (2019) offer empirical evidence of how the idea of a supposedly innate community involvement in ecotourism is challenged. Their conclusion that the levels of participation tend to be low and not directly proportional to the distance are a challenge to the traditional theoretical assumptions like distance decay. More importantly, the insignificance of the current participation in decision-making and activities that allow earning money highlights the disproportion in the structure, where local communities are placed as the passive beneficiaries but not active stakeholders. This is an indicator that community participation principle which has more often been hyped in theory is often compromised by the unbalanced power relationships and lack of capacity building.

Regarding demand-side, Patwary (2023) indicates that the environment beliefs and commitment to conservation play a vital role in determining the levels of environmentally responsible behaviour of tourists. This will set a definite behavioural course between awareness to action; however, it presupposes a certain individual responsibility, which could fail to take into account structural and contextual limitations, including poor infrastructure or lax regulation. Thus, despite a clearly defined core value of ecotourism, its practical application is extremely reliant on the quality of governance, the representation of stakeholders, and the consistency between tourist behaviour and institutional backing, demonstrating an apparent disconnect between theory and practice.

Ecotourism is often popularized as a model of sustainable development but its positive effects are more and more disputed because of the question of authenticity and practice. According to Garcia and Vargas (2024), one of the biggest issues here is that the organisations fakes their environmental responsibility by masking themselves as environmentally responsible in order to acquire the eco-conscious consumers. This brings to the fore a reputational risk to the tourism industry but equally so unveils a deeper rooted issue, of sustainability in the ecotourism sector being a marketed commodity, but not a practicable idea of how things actually are run. The fact that little has been done on greenwashing in tourism also points out that the magnitude of the issue can be underrepresented meaning that there is a gap in terms of scholarly research as well as legal control.

In a more general sense, Shasha *et al.* (2020) show that ecotourism brings financial growth, preservation, and community health, which is corroborated by the rising academic interest on an international scale. The premise that ecotourism implicitly produces ecological and socio-economic results that are balanced is rather naive in its own right since their report, in fact, reflects that there is a mix of ecological and socio-economic effects. This dichotomy is an indication that gains are very conditional, and they require effective governance and planning. Ecotourism, therefore, cannot be considered as a uniformly sustainable approach but is more of a situational policy that needs a meticulous formulation.

Just in the same way, Stronza, Hunt and Fitzgerald (2019) define ecotourism as a conceptual two-way relationship between conservation and local development as a win-win strategy. Although there is empirical evidence that it has the potential to improve biodiversity protection and diversify livelihoods, this idealised framing does not reflect the reality that there are inequalities in the allocation of benefits, and governance presents challenges that have the propensity to hamper the outcomes in reality. The lack of uniformity of findings in the various destinations shows that ecotourism may not necessarily result in the success of conservation. Rather, its success should depend on well-established institutions of governance, fair involvement of constituents, and review of evaluators. In general, despite the numerous positive environmental and socio-economic outcomes of the ecotourism industry, the recent commercialisation of ecotourism, lapse in its implementation and governance issues opens a major conflict between the planned potential and the actual picture of ecotourism.

For the present study, this literature is relevant because Hikkaduwa represents a destination where ecotourism depends on a fragile coral reef ecosystem, but also on local livelihoods, visitor behaviour and institutional management. Therefore, the concept of ecotourism is examined not only as a tourism label, but as a practical governance challenge.

## 2.3 Environmental Sustainability in Coastal Tourism

Sustainability of the environment with tourism is a concept that has been embraced massively, but there exists a big gap between promise and performance. According to Gössling and Scott (2018), although the climate change is considered a major problem in the tourism industry, there is no consensus on how and when decarbonisation should be implemented. Most of the stakeholders believe that further increase in tourism can be maintained with a reduction in emission, commonly based on technological solutions in the future. This depends on unprovable technological optimism is a crucial weakness since it postpones action and also strengthens a business-as-usual strategy instead of facilitating any transformation. Moreover, the existence of opposing belief systems and so-called agnogenesis emphasizes the fact that uncertainty is maintained in the industry and hampers the change of responsibility and the slowing-down of policy.

Likewise, Higgins-Desbiolles (2018) offers a more radical perspective on this issue by claiming that there is no way that growth-based model of tourism can fit the concept of sustainability. The article has brought out the fact that economic motives always work above the environment and even social advice resulting in overtourism and ecological destruction. The logic underlying this critique is that the current model is not structurally sustainable, and in reality, it can be argued that the fundamental assumption of tourism growth and sustainability is that the two can co-exist. In addition, the realization of identification of sustainability initiatives as superficial and marketing oriented supports a point of view in relation to the greenwashing question, that a lot of sustainability processes work more as a symbolic adherence than a change of direction. The emotion of a Degrowth based solution further highlights that the system needs to change and not to be improved gradually.

Conversely, the article by Pulido-Fernández, Cárdenas-García and Espinosa-Pulido (2019) presents research findings that the environmental sustainability can have a positive effect on tourism development, especially with well-established regulatory guidelines. However, this positive view should be taken with a grain of salt, because the specified two-way interaction makes it clear that the growth of the tourism industry causes environmental pressure to escalate at the same time. This shows that sustainable development-oriented growth might not always decrease environmental effect without very close control. Consequently, Sustainability can increase the competitiveness, but its success is greatly contingent upon the quality of governance and the capacity to strike the right equilibrium between economic growth and environmental boundaries. Taken together, the literature indicates that there is a fundamental tension between growth and sustainability, and viable and more realistic and enforceable methods need to be provided.

Coastal ecosystems are some of the most productive and vulnerable ecosystems whose degradation is increasingly being observed to occur at a rapid rate. According to the report provided by Eddy *et al.* (2021), over the last 50 years, coral reefs have lost two out of every three coral reefs in the world due to the loss of ecosystem services demanded by them since the 1950s, fish productivity has decreased by approximately 60%, and the biodiversity associated with it has decreased by two-thirds. Although these results provide a clear picture of large scale ecological deterioration they also describe a major weakness of the present management strategies, which seem inadequate to deal with cumulative demand due to climate change, overfishing and pollution.

The fact that fishing effort has been on the rise even as yields have been dwindling, further shows that there are structural inefficiencies in the system, which indicates that there is no correlation between exploiting resources and ecological sustainability. Guannel *et al.* (2016) point out that coastline security is most suitable when ecosystems like the coral reefs, mangrove, and seagrasses are operated as a single network. Their results demonstrate the great contribution of mangroves in attenuating the waves energy and coastal erosion with help of the coral reefs and seagrasses. This combined face raises the issues of reductionist management methods that do not consider these ecosystems as blocks of a bigger ecosystem which may decrease the total coastal resilience. Although the seagrasses have the potential to partially offset the destruction of corals, this is not the case that should be taken as an indication that coral destruction should not be countered by conserving the corals, but on the contrary conserving the corals would lead to normalisation of the ecosystem degradation. The ecosystem functions are also context-specific and this makes policy application more problematic and location-specific strategies need to be implemented in lieu of generalised solutions.

Reed *et al.* (2022) add to this discussion emphasizing that the effects of climate change on the foundation species, leading to the stability of the coastal ecosystem, are profound. The fact that they identified threshold responses, when little change on environmental change will be sudden collapse, proves that the environmental change is non-linear. That opposes the conservative methods of management that might not be able to predict sudden changes in the ecology and the permanent destruction. Moreover, reliance on foundation species, which is the determinant of ecosystem resilience, highlights the weakness of ecosystems. Hence, although scientific awareness advances, the fact that degradation of the ecosystem continues implies that there is a chasm between ecologically informed behavior and its proper adoption into policy in the formulation and control of multifaceted system activities.

The tourism benefits of coastal tourism are very lucrative to the economy; however, there is increasingly mounting evidence to reflect its multifaceted and usually harmful effect upon the environment. Eider *et al.* (2023) conceptualise the SCUBA tourism as a social-ecological nexus, with the patterns of tourist interventions relying on tourist preferences and business networks. Their results show that diving pressure is very concentrated in the well-known locations, which results in overcrowding and ecological distress. This level of utilization indicates a core management failure, as preferences on tourism in the markets are more important than the ecological carrying capacity.

Though other points of diving are available, inequality in distribution remains the same and demonstrates that either the governing mechanisms are weak or poorly implemented thus hindering the effectiveness of the sustainability strategies. On the same note, according to Flynn and Forrester (2019), boat anchoring is another significant but least appreciated contributor to coral reefs deterioration, covering multiple species and fish stocks have significantly decreased. The immediate physical devastation which the anchors have, along with the especially the ecological effect, proves the magnitude of this blow. Considering that such an easy to avoid, and well-reported problem still persists in such great percentage of reefs, all demonstrates a serious gap between scholarly research, and implementation of management in practical use. Additionally, the absence of the general use of such alternative methods as mooring system signifies that there is no strong regulation, enforcement, and worsens the situation of ecosystem degradation.

Moreover, Grelaud and Ziveri (2020) reveal a sharp rise in the amount of marine litter during the high tourism season, and the waste products associated with tourism constitute much of the beach pollution. Although official measures could minimise litter by more than half, the result highlights the point that environmental degradation is not a necessary process but mostly the result of poor management and control of behaviour. The excessive use of seasonal interventions however brings the issue of the sustainability of these initiatives since they might solve the symptoms as opposed to the structures that contribute to the production of wastes in the mass tourism systems.

O'Driscoll *et al.* (2019) also mention the environmental stressors that come along with coastal tourism including augmented wastewater discharge and nitrogen contamination. They identify that wastewater drainage that comes with tourism is a major cause of groundwater pollution especially at peak seasons when people visit the places. This has shown that infrastructural development has not been keeping up with tourism development and this has led to the possibility of externalities on the environment that tend to be ignored during the planning of destinations. In addition, the inconsistency of the effectiveness of the treatment implies that technological solutions will not be adequate without continuous regulation and monitoring procedures.

On a more abstract level, Smith *et al.* (2023) claim that the natural and social harms of coastal tourism are much more harmful than the economic one, although the economic advantage is often prioritized. Their results note that there is rampant environmental degradation and inequities in society with economic benefits being unevenly distributed. This dismantles the priority given to tourism as a means of sustainable development, and instead of underscoring its contribution towards systematic inequalities in the world and environmental abuse. Moreover, lack of integrated approaches to the coastal zone management implies that governance is disjointed, and it cannot respond to compound and interdependent effects appropriately. The economic aspect of the coastal tourism development is notable but the environmental one is considerable and frequently worsens by the inappropriate governance, a lack of infrastructure, and pressures of the market, which opens the gap between the rhetoric of sustainability and reality of operation. This literature is directly relevant to Hikkaduwa because the destination depends on beaches, coral reefs and marine tourism activities that are environmentally sensitive. The coastal tourism literature therefore helps explain why unmanaged tourism pressure, waste, wastewater, erosion and reef damage are central to the sustainability of the case study area.

## 2.4 Environmental Policy and Governance in Tourism

In terms of environmental management, tourism policies are aimed at controlling the impact of the industry, integrating development with sustainability, and reducing the risk associated with climatic changes and conditions, although, their performance is not really consistent. Becken *et al.* (2020) indicate that the overall integration of policy accurately reflects between tourism and climate change is rather low, as most frameworks are climate driven, as opposed to tourism-driven. Tourism is mostly positioned as a weak industry and little is done to appreciate the contribution of the activity towards emissions. This is an oversight in a critical policy, in which tourism comes out as a passive victim of climate change but an active source, in order to restrict responsibility and mitigation measures. Moreover, absence of specific goals, quantifiable targets, and intersectoral alignment means that most of the policies are not implemented. This reveals that the environmental policies tend to act as strategic statements rather than action tools, which is put to the fore of their weakness in being practical. On the contrary, Miccini, Ciappei and Liberatore (2026) indicate that regulation interventions may be successfully used to lessen the intensity of tourism, especially when policies are directly relevant to tourism.

Their results prove that the visitor density decreases with the pressure of regulation giving new significant importance to the role of governance in regulating overtourism. The weak performance of the indirect policy including tourist taxation indicates that it might not be effective enough to implement significant changes in behaviour through the working of market-based instruments. Furthermore, the dependency on regulation mostly when there is a high tourism pressure demonstrates a reactive instead of a proactive policy stance, which might prove to be poor in dealing with long-term sustainability issues. Schönherr, Peters and Kuščer (2023) also demonstrate that a crisis like the COVID-19 can serve as a trigger to empower the policies aimed at sustainability, raising awareness and enhancing the use of integrated policy tools. Although this shows the resilience of tourism governance to change, it also shows that tourism governance has a dependency on external shocks to initiate a meaningful policy response, as opposed to long-term commitment to sustainability objectives. The application of environmental, economic, socio-cultural, and communication tools suggests a wholesome policy structure but the usefulness of any such action is at the end of the day hinges on the implementation, enforcement and alignment of the stakeholders which is not always evenly matched across the destinations. In general, environmental policy is relevant in driving sustainable tourism, but its effectiveness is limited by the lack of proper integration, reactive formulation, and gaps in implementation, all of which have raised the issue of lack of congruency between policy purpose and its actual results.

Tourism policy implementation is a complex process that is conditioned by the multi-level governance systems, stakeholder relationships and limitation in situations. According to Bichler (2021), the tradition of tourism governance is predominantly dominated by both the government and other non-governmental actors, where the residents of the local areas have never participated in making decisions. Although the classification to determine the level of participation such as limited to inclusive gives evidence of a cumulative move towards a more participatory form of governance, the implementation of the same is almost purely theoretical since there are still institutional and institutional constraints and lack of resources needed to practice the level of participation. With all the improvements in sustainability and legitimacy, more participation, the assumption that the inclusion will translate to more, does not consider power imbalances and the capacity limitation of the local populations hence participation alone cannot work without structural assistance.

In the same vein, Guo and Li (2025) underscore the relevance of multi-level governance and adaptive governance in the implementation of the low-carbon tourism policies. Their results prove that not all policies are effective in every geographic area, because of the state of the economy, the ability to govern, and environmental forces. Though the fact that the implementation paths are diverse (technology-driven, community-based and policy-led) provides flexibility, the variability also means that there is no standardisation, and thus, inconsistent policy outcomes and performance on sustainability across destinations are also possible. Moreover, the use of context-specific strategies, which is required, makes scaling and transferability of policies, which limits the applicability of successful patterns to a wider range.

Roxas, Rivera and Gutierrez (2020) also support the significance of the stakeholder collaboration, assuming the picture of the tourism governance as a chain of values that encompasses governments, business, communities and tourists. Although this framework is a valuable way to give a full picture of the shared responsibility, it arguably underestimates the problems of aligning various stakeholder interests, especially in the situations when the relative powers and orientation of priorities are conflicting. It is conceptually important that tourists should be regarded as active stakeholders, although, in fact, this viewpoint can make them appear more influential than they can actually be since the individual behavioural change is often restricted by the systemic and infrastructural constraints.

Scuttari *et al.* (2025) propose the notion of a science-policy gap, which refers to a gap between the production of the knowledge and its practical use. The results of their study indicate that participatory and adaptive styles may improve the use of knowledge; yet, the further focus on data gathering rather than acting is a sign that it does not generate action and refers to the institutional inertia and poor governance systems as the major hindrances. This shows the necessity to do better translations of research in policy. Thomann, van Engen and Tummers (2018) highlight the importance of frontline workers that proves the difference in the perceived discretion and policy meaningfulness in terms of willingness to implement an intervention. Although this holds true to bottom-up implementation theories, it also demonstrates a significant weakness: the lack of proper resources, organisational support, and clear policy direction will not allow the individual motivation to be effective to ensure implementation.

All this implies that the dynamics of policy implementation in the tourism industry include interlaced relationships between the systems of governance, the involvement of stakeholders, and the legitimacy of the institutional capital, and the fact that the lack of implementation keeps playing out indicates that the current theoretical developments have not yet been applied into regular practical frameworks.

Such implementation gaps in environmental governance raise the issue of the continuing disproportion between intentions in policy formulations and their realization in tourism situations. Haseeb and Azam (2021) prove the existence of a close connection between tourism development, the quality of governance, and environmental pollution as both higher tourism and higher levels of corruption increase CO2 emissions. Although democracy is realized to have curbed the effects of the environment in more-income settings, it has not been effective enough in the low-income nations, which portrays institutional inefficiency. This implies that the quality of governance, and not the presence of policies, determines the environmental performance, throwing a doubt on the belief that sustainability policies directly lead to better environmental performance. In addition, the two-way interaction between tourism and corruption means that the development of tourism can be active in strengthening inefficiency in governance, as opposed to being influenced by them, which makes implementing the policy more difficult.

This criticism is further extended by Jamal and Camargo (2018) to state that tourism governance is inherently influenced by neoliberal, market-based interests based on which the process and priorities of economic development are prioritized over equity and sustainability. Their results point out how local communities are not part of the decisions they make which leads to inequality and even unfair outcomes. This points to a structural weakness in governance regimes, in which the processes of participation are symbolic, and not real, transferring no power nor remediating institutional inequalities. Also, the emphasis on procedural inclusion and not substantive results implies that governance systems might emphasize legitimacy in comparison with the genuine social and environmental justice thus solidifying prevailing inequalities.

Ramukumba (2021) lays stress on the stakeholder involvement and co-management in ensuring effective environmental governance of protected areas. Although the research shows that joint solutions have the potential to increase conservation success and community assets, it also takes into account traditional trends of removal and displacement. This points to a contradiction in conservation based tourism where the sustainability efforts have played a concurrent role in environmental conservation and social discrimination. Despite being established as a plausible solution, co-management seems to be largely a context-dependent concept, and it can be successful only in the case of transparency, trust, and sharing benefits equally, which is not always easy to realize in practice.

Sharpley (2023) offers the more generalized critic by suggesting that sustainable tourism governance has not achieved much even with the mass support of the policy. The dominance of growth-oriented agendas still gives precedence to the need to boost the number of tourists to environmental capacity. This unveils a major paradox of tourism governance in that sustainability is a rhetoric goal that is advanced whereas growth is the main goal in reality. In addition, the identification of the tourism as a global system implies that local governance is not enough. This puts the efficacy of the destination level policy in question, suggesting that in absence of coordination at global level, sustainability initiatives will be too fragmentary and irrelevant. Taken together, weak institutions, power imbalance, and prioritization can be identified as the factors that create implementation gaps in the sphere of environmental governance, and the fact that these issues still persist suggests that the attainment of sustainable tourism will require systemic changes, rather than a series of appearance of new policies.

The stakeholders have been found to be at the centre of converting the environmental policies into practice; the stakeholders are influenced by the dynamics of governance and boundaries of institution. Bichler and Lösch (2019) also assert that collaborative governance can work only with the involvement of a leadership engine that will enable the coordination process, trust, and interactions with the stakeholders. Although effective collaboration relies on strong leadership, the research also points out that uncertainty and lack of confidence in stakeholders are among the results of institutional changes and weak leadership. This implies that collaborative governance is a highly contingent matter on personal and institutional stability that can easily be disrupted and its sustainability is also quite short-lived.

Moreover, the partiality of locals to governance implies that the governance processes are selectively inclusive. This brings in fears that there might be more procedures over collaboration that may not be actually inclusive to redistribute the authority of making decisions but rather strengthen the preexisting hierarchies of power.

As Loverio, Chen and Shen (2023) establish, stakeholder collaboration can adequately deal with overtourism by degree of coordinated measures like visitor control and capacity management. Though this demonstrates the practical importance of collaboration, the ingrained nature of the overtourism phenomenon by outside forces like social media reveals that the activities of local stakeholders might not be effective enough to regulate the demand of tourism. Moreover, the residents continue to have positive attitudes because of the economic benefits added but this can conceal the potential social and environmental costs in the long term meaning that there will be a possible trade-off between the short-term and sustainable results.

Nyaupane, Poudel and York (2024) also add that no single model of governance can be able to deliver conservation, livelihood and tourism goals at the same time. Their results advocate the importance of the polycentric and adaptive systems of governance. The multi-layeredness of governance systems can also create coordination issues that will delay the decision-making process and make the policies less efficient. Furthermore, adaptive co-management lays stress on the high level of institutional capacity and unceasing stakeholder involvement, which are not always possible within the context of limited resources, which restricts applicability in practical settings.

Using the relevance of stakeholder collaboration, Wondirad, Tolkach and King (2020) also establish the significance of this collaboration especially in the developing countries where a lack of a well-developed coordination and mistrust result in ineffective ecotourism realization. The results of their research point out that ineffective collaboration may lead to the deterioration of the environment and unequal distribution of benefits. This shows that stakeholder collaboration even when it is being advocated in a massive way, is not naturally good and can fail without clear structures, accountability systems and capacity building. Moreover, the assumption that everything works out as it should and both parties share similar motivations can ignore the involvement of conflicting interests and power imbalances among the parties involved.

This governance literature is relevant to the present study because Hikkaduwa is shaped by multiple institutions, including tourism, coastal conservation, local government and protected area management bodies. The literature therefore supports the study's focus on the policy–practice gap and the institutional challenges that affect implementation at destination level.

## 2.5 Chapter Summary

This chapter critically reviewed the literature on ecotourism, environmental sustainability, coastal tourism impacts, and environmental governance. The review highlighted that although ecotourism and sustainability are widely promoted as policy objectives, their practical implementation remains limited due to governance weaknesses, inadequate stakeholder coordination, and persistent policy–practice gaps. The literature also emphasised the environmental pressures associated with coastal tourism and the challenges of managing fragile ecosystems effectively. These findings indicate that the key issue lies not in the absence of policies, but in their implementation at the destination level.

# CHAPTER THREE

## 3.1 Introduction

The methodology adopted to investigate implementation of ecotourism and environmental sustainability policies of the Hikkaduwa coastal locality of Sri Lanka is presented in this chapter. It provides the rationale for the research philosophy, approach, case study strategy, methodological choice, time horizon, data collection methods, sampling strategy, data analysis process, ethical consideration and limitations of the study.

The design of the study is a qualitative case study that is followed by Policy Document Analysis and Semi-structured interviews with the key stakeholders. Because these methods would enable the study to explore both the formal policy intentions as well as the lived experience of the stakeholders in context, they were chosen. The policy document analysis indicates official policy framework, management mechanisms and institutional responsibilities in relation to the given policy document; the interviews provide inputs regarding the experience of the policy in practice. In this chapter the research design that was created for the research aim and research questions is described in a systematic and credible manner.

## 3.2 Research Philosophy

Research philosophy is the set of assumptions that shapes the understanding of reality and knowledge production in research. In this study, an interpretivist philosophy is taken up. According to Fischer (2025), interpretive inquiry is not interested in determining or assuming that social and policy realities are fixed or value-free matters, but how people act based on their understanding of those realities. Equally, Cena *et al.* (2024) observe that the interpretivist epistemologies fit suitably in cases where the research aims to consider the meaning, experiences, and socially constructed understandings as opposed to objective measurements alone. The significance of these ideas is that whereas this study is not merely interested in the formal existence of the ecotourism and environmental sustainability policies but on how the policies are conceived and felt in practice.

In an interpretivist stance, the creation of knowledge is based on the interactions with the views of the participants, and their contexts of view formation. According to Scheibelhofer (2023), qualitative research orientation toward interpretations is aimed at introducing the data, which can reflect the own meanings and understandings of the participants. Similarly, Aprimadya (2025) demonstrates how the interpretive approach can help us explore implementation as an ever-changing process that is influenced by norms, activities, and context, as opposed to a technical exercise. This is particularly applicable to studies relating to policy implementation, where various actors can have different interpretations of the same policy.

This philosophy is appropriate for the current study because the purpose of the study is to explore stakeholders' perceptions and experiences on the implementation of ecotourism policy and environmental sustainability policy in Hikkaduwa. The study is not designed to statistically measure the implementation of the policy. Instead, it is about meanings, perceptions, and institutional roles and experience in these contexts. For the reasons mentioned above, an interpretivist approach to philosophy is qualified for this study; it allows for an in-depth grasp into the understanding and practice of policy.

## 3.3 Research Approach

This research is done using an inductive research approach. The general inductive approach, as described by Locsin and Schoenhofer (2024), enables such a reduction of the raw qualitative information into meaningful categories and establishing explicit connections between the research goals and the results. On the same note, Robinson and Bailey-Rodriguez (2026) claims that inductive research encourages the exploratory formulation of propositions, models, and conceptual frameworks, not that based on a set of propositions to be tested. It is also mentioned by Barrett and Younas (2024) that inductive reasoning is typically connected with qualitative research as it transitions between particular observations and narratives of the participants towards general patterns and interpretations. A combination of these perspectives indicates that we should induce the data in a case when the goal is to learn not to apply a known theory to the data but to learn what it teaches us.

This is an exploratory study, making the inductive approach appropriate. Does not test a predetermined hypothesis Rather, it seeks patterns, themes, challenges and implementation gaps in policy documents and from stakeholder interviews. This enables the findings to be revealed from the evidence collected in the study in Hikkaduwa and to encourage understanding at destination level about the policy–practice gap.

## 3.4 Research Strategy: Case Study

In this research, a qualitative case study approach is taken. Mtisi (2022) explains that qualitative case study research is better to investigate complex social phenomena in context and to rely on various evidence-gathering sources to comprehend them profoundly. Chowdhury and Shil (2021) also believes that case study research is good since it encapsulates the holistic and meaningful nature of practical events instead of making them disjointed with their conditioning environment. Lim (2025) further adds in the same line that qualitative research can be very helpful when one is seeking to come up with deep human understanding of issues that are complicated. Combined these arguments indicate a qualitative case study is suitable when the researcher aims at an in-depth understanding on how a process is applied in a particular environment.

The case that will be examined in this study is that of Hikkaduwa coastal area in Sri Lanka. It is not what is declared in the policies of ecotourism and environmental sustainability, but rather how it is understood, perceived and the actual practice by the concerned parties. The use of a case study approach is thus appropriate since it enables the researcher to analyze the policy as it occurs in a practical framework which covers policy documents as well as interviews with the stakeholders.

Hikkaduwa is chosen as it is a very significant case to this research topic. It is a tourism destination that is located on the coast and environmental protection, tourism development and governance issues collide. This renders it an appropriate environment to investigate the way policy functions on the ground, what issues are occurring where there is implementation, and why loopholes might exist between policy intentions and on-ground realities. The most appropriate approach hence is the qualitative case study since it enables one to gain an in depth insight on the policy implementation in its real world setting, and not as an abstract issue.

## 3.5 Methodological Choice

In this research, this study adopted a qualitative methodological approach. According to García-Fernández *et al.* (2025), qualitative research is useful as it helps record experiences, perceptions, and meanings that would otherwise be overlooked in quanitative research. Similarly, as Bingham (2023) remark, qualitative inquiry is oriented in reaching out to rich textual information and facilitating close reading as to patterns, opinions, and context.

The research concentrates on meanings, experience, and the views held by the stakeholders, as opposed to testing numerical relationships. Given that both access to policy document and access to interview are involved in producing the qualitative data, having a single coherent qualitative methodology choice assists in sustaining the depth, consistency and a clear focus in the research.

## 3.6 Data Collection Methods

This research adopts both secondary research and primary research methods in analyzing the application of ecotourism and environmental sustainability policies in Hikkaduwa. Both data sources are essential to have a thorough insight into the policy intentions and their realization in practice. Secondary data provide the formal policy and regulatory framework whereas primary data offer practical information of stakeholders that are directly involved with policy implementation.

### 3.6.1 Secondary Data

Secondary data used in this analysis was official policy documents, government reports and tourism strategy publications in ecotourism and environmental sustainability in Sri Lanka. The selected documents included the Sri Lanka Tourism Development Authority (SLTDA) Strategic Action Plans, National Sustainable Tourism Certification guidelines, and the existing tourism development reports. Also, there are environmental and coastal management documents that include the Coastal Zone and Coastal Resource Management Plan (2024-2029) by the Coast Conservation and Coastal Resource Management Department. The documents by the Ministry of Environment and National Audit Office concerning coastal and environmental governance are also supported. These documents are chosen as they are directly related to tourism management, environmental management and coastal management at Hikkaduwa. To be reliable and authentic, all the documents are retrieved on official government websites and within institutional sources. To reflect the current policy frameworks and implementation practices, only recent and relevant documents are used.

The process of data collection entails a systematic identification, selection and review of each document. Topics covered in these sections are analyzed to pull out information on policy goals, implementation processes, institutional role and regulatory processes. The data obtained are classified under certain frameworks including governance structure, enforcement practices, sustainability strategies, and stakeholder roles to facilitate systematic analysis. The secondary data assists in giving an accurate idea of the official policy framework, such as what policies are in place, how they are designed, and what they aim to accomplish. This enables the research to have a starting point in terms of real policy content and then proceed to examine how these policies are put into practice.

### 3.6.2 Primary Data (Interviews)

In this study, primary data was gathered by conducting a semi-structured interview with four participants chosen to take part in the process of tourism governance and implementation of environmental policies in Hikkaduwa. The stakeholders are one officer in Sri Lanka Tourism Development Authority (SLTDA), one officer in the Coast Conservation and Coastal Resource Management Department, a representative of the Hikkaduwa local government authority and one local boat operator involved in coastal tourism activities. The desired participants are purposively sampled in accordance with direct participation in policy implementation, regulation and decision making.

Participants were identified through institutional sources and professional networks. They were contacted by email, and each participant received clear information about the purpose of the study before the interview. Consent was obtained before each interview. The interviews were conducted online, depending on participant availability.

The interview guide is structured in such a way that the same pattern is adopted in all interviews. The questions are specific on the policy implementation processes, roles and responsibilities of agencies, interagency coordination, enforcement practices, and implementation challenges. They present in-depth answers; depending on the experience of the participants and their participation in the sector.

| Participant Code | Stakeholder Type | Reason for Inclusion |
|---|---|---|
| P1 | Sri Lanka Tourism Development Authority officer | Represents tourism policy and destination-level tourism governance |
| P2 | Coast Conservation and Coastal Resource Management Department officer | Represents coastal conservation, permits, and environmental regulation |
| P3 | Hikkaduwa local government representative | Represents local service delivery, waste management, and local governance |
| P4 | Local boat operator | Represents local tourism operations and practical experiences of tourism activity |

*Table 1 Profile of Interview Participants*

Participant codes were used to protect the identity of interviewees while allowing the findings to distinguish between different stakeholder perspectives.

The interviews were conducted online, depending on participant availability. With the consent of participants, the interviews were audio-recorded for accuracy. The recordings were then transcribed into written text with the assistance of AI transcription tools. After transcription, each transcript was manually checked and edited by the researcher against the interview recording and interview notes to ensure accuracy and preserve the meaning of participants' responses. The transcripts were anonymised using participant codes before being used for thematic analysis.

Purposive sampling was utilized in order to choose the participants directly involved in the tourism governance and implementation of environmental policy within Hikkaduwa. These four participants are as follows: one officer of Sri Lanka Tourism Development Authority (SLTDA), one officer of Coast Conservation and Coastal Resource Management Department, one representative of the local government authority of Hikkaduwa and one local boat operator. The reason behind the choice of these individuals is that, their functions entail in policy planning, regulations and implementation within tourism and environmental sectors. The selection criterion to be followed is to select participants owning current knowledge, work experience and responsibility to decide on ecotourism and environmental sustainability policies. The participants were selected using institutional sources and professional networks to ensure that the data were informed, relevant and credible.

Although the sample size was small, the study did not aim to achieve a broad or statistically representative interview sample. Instead, it focused on selecting one representative from each important governance level or stakeholder sector connected to tourism and environmental policy implementation in Hikkaduwa. These included tourism regulation, coastal management, local government service delivery and local tourism operation. This approach allowed the study to explore, in greater depth, how different stakeholders experience policy implementation and governance challenges from their own institutional or practical positions. Therefore, the value of the interview sample lies not in its size, but in its ability to illustrate different perspectives on the policy–practice gap in Hikkaduwa.

Purposive sampling would be appropriate in this research since this study requires elaborate experience-based information on the perspective of those directly involved in the implementation of these policies. The sampling should not be random, since it can contain the participants without having the associated information. The study chooses participants who are experts and relevant which further ensure that the data gathered is focused, meaningful and which are also directly relevant to the study aim and questions.

## 3.7 Data Analysis Method

Policy documents and the interview data were analysed using thematic analysis adopted (Braun and Clarke, 2006). This approach was suitable in this study to find out repetitions, problems and message in the implementation of the policy in Hikkaduwa. The analysis concerned itself with formal policy intentions compared to the experiences of the various stakeholders with implementing the policy as it manifested on the ground.

The policy documents and the interview notes which had been chosen were carefully written down and analysed using the qualitative content analysis software program “Taguette”. The analysis started with familiarization of the documents and the answers of the interviews. Subsections of text for each of the following key concepts related to policy, institutional role, permits and standards, coastal management, monitoring, coordination, resources, tourism pressure and visitor behaviour, and waste and wastewater and reef damage, were then coded.

For each one of those relate codes more general codes were gathered under those codes. The four main themes emerged were: sustainable tourism governance; challenges in implementing policies; tourism behaviour and livelihood pressure; and coastal environmental degradation. These themes were used to organise the findings in Chapter 5. Both the document analysis and the results from interviews were analysed according to a common thematic structure which enabled comparison to be made between formal policy mechanisms and the practical experience of stakeholders.

The analysis was based on a step-by-step thematic process. The selected policy documents and interview notes were read several times to become familiar with the content. The relevant sections were coded for issues including the goal of the policy, the institutional role, enforcement, coordination, monitoring, the pressure from tourist numbers, visitor behaviours, waste management, wastewater and reef damage. Codes were similar codes were lumped together under more general headings. These categories were subsequently translated to 4 main themes: sustainable tourism governance, problems with policy implementation, tourism behaviour and livelihood pressure, and degradation of coastal environment. For both policy documents and interview notes a common theme was used to ensure that formal policy intentions could be related to stakeholder experiences

In doing this process, the relationship between the policy intention and the implementation reality were identified. It also facilitated the second objective of linking research findings directly with the research questions, including the policy issues, problems of implementation in real-life contexts, and policies and factors affecting policy effectiveness.

## 3.8 Researcher Positionality

The present study was conducted with an interest to know how ecotourism and environmental sustainability policies are practically implemented in the coastal tourism destination. Prior to the study, I was expecting that there would be well-established tourism and environmental policies in place in Hikkaduwa, and there might be some difficulties in enacting them in practice. I knew that my own views about the implementation of policy and protection of the environment may have shaped my interpretations of the data. So I tried to be reflective throughout the research as much as possible, for example, I checked out policy documents which were relevant to the interviews, and I let themes come out of the data, rather than imposing them myself.

## 3.9 Ethical Considerations

Ethical considerations are carried out well in the course of data collection and analysis in this study. Informed consent is taken before every interview by all four interviewees. All participants were given a good explanation on the intended use of the research, the nature of the questions to be asked and the use of the data. Inclusion is confidential and the informants are told that they are free to say no to any question, and quit the study at any point without any repercussions.

To protect participants, no identities of the participants are being revealed in the study to preserve confidentiality. Rather than giving names to their participants or act-utilizing specific job descriptions, the participants are identified by generic identifiers (Participant 1, Participant 2, Participant 3 and Participant 4). Any sensitive information provided in the interviews is treated with extreme caution and is not reported in such a way as to raise identity of individuals or institutions.

The responses to all the interviews are written down in form of documents and are kept in safe places. The researcher is the only person that has access to the data. The data is utilized specifically to this research in an academic sense. These ethical considerations are to make sure that the research process is responsible and to safeguard the rights of the participants, and to uphold the research integrity and credibility.

## 3.10 Limitations of the Study

There are a number of limitations around this study. The interview sample was small and consisted of four participants. This narrowed the scope of stakeholder opinions, however, the stakeholders selected were purposefully chosen, so that they represent, inter alia, various stages of policy outreach or implementation: tourism regulation, coastal management, and service delivery/policy implementation by local authorities and local tourism operations etc. So it is not the level of representation in the interviews, but rather the depth of the lived experience of all the stakeholders involved, juxtaposing the institutional view with the local view. Due to time and availability, extra stakeholders were not able to be accessed. More local tourism operators, environmental NGOs, rangers, and a Visitor Information Officer (VIO) or local business association could also have been included in the study.

The interviews that were available, however, help provide valuable perspectives on how the policies are implemented and lived by those directly involved in the lives of tourism and coastal management stakeholders in Hikkaduwa. The study does not claim to represent all stakeholder views in Hikkaduwa. Instead, it provides an exploratory and in-depth understanding of policy implementation through selected key stakeholders directly connected to tourism regulation, coastal management, local governance and local tourism operation.

Policy documents and official reports were also collected for the study as secondary data. These documents help to clarify formal policy intentions, but they don't necessarily describe how policies are implemented. This restriction was mitigated by comparing the results of the document analysis with the results of the interviews. The study is undertaken in a particular case study area (Hikkaduwa) and hence the results may not be directly applied to all the coastal tourism destinations in Sri Lanka. The case offers some interesting insights on destinations-level policy implementation challenges, though.

## 3.11 Chapter Summary

The methodology adopted in the present study was been explained in this chapter. It explained the interpretivist stance of research, the inductive process, the qualitative case study approach, the methods used for data collection, the sampling process, the thematic analysis process, the ethical considerations and the limitations of the study. The chapter demonstrated the suitability of using policy document analysis and semi-structured interviews methods in analyzing the implementation of the policies in Hikkaduwa relating to ecotourism and environmental sustainability. The methodology enabled the study to examine the formal policy intent with the experiences of the stakeholders and thematise directly relevant to the research questions.

# CHAPTER FOUR

## 4.1 Introduction

This chapter provides the contextual background necessary to understand the case study of Hikkaduwa. It outlines tourism development and environmental policies in Sri Lanka, followed by a discussion of the Hikkaduwa coastal area, including its tourism system and environmental challenges. This contextual overview supports the analysis of policy implementation presented in later chapters.

## 4.2 Ecotourism and Environmental Policies in Sri Lanka

This section provides only the national background to tourism, ecotourism and environmental policy in Sri Lanka; the detailed analysis of specific policy documents, mechanisms and implementation issues is presented in Chapter 5. The economic development of Sri Lanka has been boosted especially in the post war era where tourism has contributed foreign exchange as well as employment and national development. Fernando (2017) emphasizes that even though natural and cultural tourism is strong in the country, the traditional performance there has been low because of irregular policies, political unpredictability, and no strategic focus. Although the 2009-touristic boom is a sign of the fast growing business, it is arguably rather reactive, looking at the new changes and developments happening and begs the question whether the tourism industry is sustainable and resilient in the long-term. Moreover, the structural issues like poor infrastructure, shortage of skilled workers and poor data systems have long persisted which implies growth in the economy has not been accompanied by any capacity building in the institutions, thus limiting the sector to facilitate sustainable tourism practice.

Kanagasingam and Rajeshwaran (2022) also reveal the weakness of the tourism industry in Sri Lanka in reference to the COVID-19 effects that resulted in severe losses of revenue, withdrawals of jobs, and the destruction of local supply chains. These findings illustrate that communities that relied on tourism were greatly affected socio-economically and it reveals how the sector is dependent on external demand. Such over-reliance on the world tourism markets displays a weakness in the structure, economic gains are quite vulnerable to the shocks taking place in the world, making the sustainability of such economic gains unsustainable. Also, the fact that the small and medium-sized enterprises have been affected unequally shows that not all development of tourism can be equally distributed and the smaller stakeholders are at higher risks when faced by a crisis.

Conversely, Sri Lanka Tourism Development Authority (2023) indicates that the performance of tourism has been high and the arrivals, foreign exchange earnings, and employment have significantly improved. Although these numbers show that the country experiences positive economic dynamics, the focus on numerical indicators of economic growth like the number of visitors and revenues could be confusing about the prospective environmental and social strains related to the growth of tourism activity. In addition, the high concentration of tourists in certain areas may also create dependency of the market creating a possibility of limited diversification, thus high susceptibility to future shocks.

Tourism development in Sri Lanka shows its high potential in the economy; however, the growth process is marked by the structural weaknesses, sensitivity to the external shocks, and tendency to focus on the growth, and not on the sustainability of such approaches, which points to the need to develop well-carried practices and more strategy-oriented approaches to the rising community.

Wijerathne, Bandara and Athukorala (2023) also emphasize that this growth of tourism in the environment has also led to erosion because of changes in land-use increasing the erosional intensity, and the ecological diversity decreased in the coastal areas like in Hikkaduwa. More organized places such as Bentota, on the contrary, have better relative results. Bentota is another coastal tourism destination in Sri Lanka, where coastal tourism development has been more organised through stronger planning and management controls. Such comparison proves the significance of strategic planning and it means that the degradation of the environment is not predetermined but is mostly due to bad governance and failure of the regulations.

## 4.3 Ecotourism and Sustainability in Hikkaduwa

Hikkaduwa is a coastal tourism destination in the Galle District, about 100 kilometres south of Colombo. It developed from a fishing village into a beach resort with hotels, guesthouses, restaurants, dive operators and glass-bottom boat activities. Its ecotourism is mainly linked to the Hikkaduwa National Park coral reef (De Silva, 1997). The Hikkaduwa Coastal Area consists of one of the most important ecotourism sites in Sri Lanka, the environment quality and tourism demand are closely linked to each other. Jayaratne *et al.* (2023) establish that coral reef condition is the most important variable that impacts the visitor preferences and willing to pay, as poor conditions result in decreased tourist interest in the reefs.

This shows the economic worth of the ecosystem health, but also points to an urgent reliance of tourism demand on the degree of ecological health, so further ecological decay presents a direct threat to the economic sustainability of the destination. Despite its potential to find solutions, as far as coral restoration is mentioned, it can be reactive and thus treating the symptoms of degradation, but not the root cause of degradation, including over-tourism and poor management of the environment. Additionally, the emphasis that is put on safety infrastructure suggests that the visitor satisfaction is multi-dimensional, but the focus on improving the infrastructure at the cost of the ecological degradation would give an opportunity to promote unsustainable tourism development. Thilakarathne *et al.* (2023) also add weight to the vulnerability of the marine ecosystem in Hikkaduwa by finding that there is a high rise in sea surface temperatures (SST), which is higher than the global and regional averages. The result is indicative of the fact that environmental sustainability in Hikkaduwa is not exclusively a management issue of local concern, but also influenced by the global climatic processes, which is problematic when developing policies and constraining the range of magnitude of local intervention. Furthermore, the rising rate of occurrences of thermal stress implies that current conservation measures might not be adequate to handle the growing rate of climate effects, which does not bode well in the long-term stability of the ecosystem.

A behavioural side of the discussion is offered by Jayasekara, Rajapaksa and Gunawardena (2024), which reveals that a pro-environmental behaviour (PEB) and motivation can significantly impact the perceived value of ecotourism experiences and environmental knowledge alone do not have a significant effect. This questions the popular view that interventions which are based on awareness are enough to generate sustainable behaviour, points which means that more intensive behavioural and motivational elements have to be considered. Individual change in behaviour might not address structural problems including the lack of regulation and environmental control which are more definitive in determining sustainability outcomes. There is an indication that Hikkaduwa is prone to the limitations of environmental integrity to achieve the potential of ecotourism, but accumulated pressures of climate change, behavioural constraints, and managerial inefficiencies demonstrate that the sensitive area is vulnerable to failure unless integrated and active measures are implemented to enhance sustainability.

To add on to this, Mahagamage *et al.* (2025) also find that microplastic contamination is rampant in coastal waters, such as Hikkaduwa, which is attributed to urban runoff and fishing. Microplastics at all the sampled sites point to environmental degradation on a large scale. This points to an important drawback of the existing management strategies, where sources of pollution are widely spread and hard to control, where cross inter-sectoral and cross-core actions will need to be taken instead of single policy responses. Moreover, there is no explicit correlation between water quality indicators and microplastic abundance, which leads a person to think that conventional monitoring systems are not effective enough to identify new environmental hazards, restricting possible responses.

According to Nisansala, Jayasinghe and Ranwella (2021), the crucial contributors to environmental degradation in Hikkaduwa include the presence of anthropogenic activities, specifically the unplanned development of tourism and the growth of coastal infrastructure. According to their results, they have strong effects on the coral reefs, the stability of the shoreline, and coastal landscapes. Herein lies a basic paradox in the development of coastal tourism where economic growth is a direct contradiction to the ecological resources that tourism relies on. In addition, the fact of unplanned development taking place further suggests poor adherence to a policy of coastal management, which strengthens the dissimilarity between planning paradigms and actualities.

Weerasekara and Illeperuma (2024) also note that the southern-western coastline is becoming more vulnerable and that the process is changing to erosion instead of accretion and covering most of the coastal aspects. Although natural occurrences and activities like tsunami of 2004 produce these changes, human beings have also been a major cause. This illustrates that the degradation of the coasts is not purely a natural occurrence but it is greatly contributed by the human induced pressures which cannot allow the narratives to deal solely with natural causes of environmental degradation. Also, it has been noted that the erosion continues even in the presence of the existing management controls, which reveals that the current interventions are too proactive, or responsive to the dynamics of the coastal changes. The elements leading to the environmental problems in Hikkaduwa are fuelled by the interaction of the failures of governance, lack of control in the development, and the emergence of the new pollution issues, and the continuation of these problems reveals the apparent lack of coordination between the environmental policy frameworks and the practical implementation.

Silva *et al.* (2022) specifically highlight the unplanned development of the tourism industry in Hikkaduwa, which, in turn, is one of the main causes of environmental destruction and decreased tourist satisfaction. Among the problems that have been highlighted in the study include the destruction of coral reefs, poor urban development, and poor infrastructure all of which have affected the sustainability of the destination. This supports the view that environmental degradation does not arise just because of increase in tourism, but due to unsustainable planning and lax regulations. As much as the suggested measures like enhanced spatial planning and architecture have created a channel through which sustainability can be achieved, their success is subject to whether they are reinforced and the stakeholders adhere to them, which has been a weak aspect in the context of Sri Lanka. In general, the current literature shows that sustainability is a well-known concept in Hikkaduwa, but there are still numerous gaps in governance and implementation that prevent the shift of the willingness to sustainability in practice.

## 4.4 Chapter Summary

This chapter provided contextual background for the Hikkaduwa case study by outlining Sri Lanka's tourism and sustainability context and the destination's coastal tourism importance. It highlighted reef-based attractions, livelihood dependence and environmental pressures, including reef degradation, erosion, waste and unplanned development, establishing the basis for the policy and interview findings.

# CHAPTER FIVE

## 5.1 Introduction

The results of the study in the implementation of ecotourism and environmental sustainability policies in the Hikkaduwa coastal area is presented in this chapter. The findings are drawn from two sources of evidence: from the policy documents and from semi-structured interview data. Document analysis was employed to look at the formal policy framework, sustainability related issues, management mechanisms and institutional responsibilities pertinent to Hikkaduwa. The interview information has been utilized to gain information on perceptions of these policies and mechanisms when experienced in practice by stakeholders within the tourism, coastal management, local governance and boat based tourism sectors.

The chapter is organised into two main parts. In the first part, results of the analysis of the policy document are presented. This section will highlight policies and supporting documents that have identified environmental sustainability issues in Hikkaduwa namely coastal erosion, degradation of reef, pollution, lack of planning, pressure from tourism and coastal developments. It also looks at how these issues are addressed, for instance through permits, setback standards, environmental assessments, coastal management plans, standards and certification etc. This chapter is focused on Research Question 1, that is, what are the-existing ecotourism and environmental sustainability policies for Hikkaduwa coastal area?

The interview results are reported in the second part. In this section, experiences of policy implementation on the ground are explored. It emphasizes on governance, monitoring and coordination, resources constraint, tourist behaviour, business pressure, livelihood dependency, waste, and wastewater & coastal environmental degradation. The corresponding Research Questions 2, 3 and 4 are linked to this part, which chiefly deals with issues of implementation in practice, institutional and stakeholder issues and factors influencing policy effectiveness.

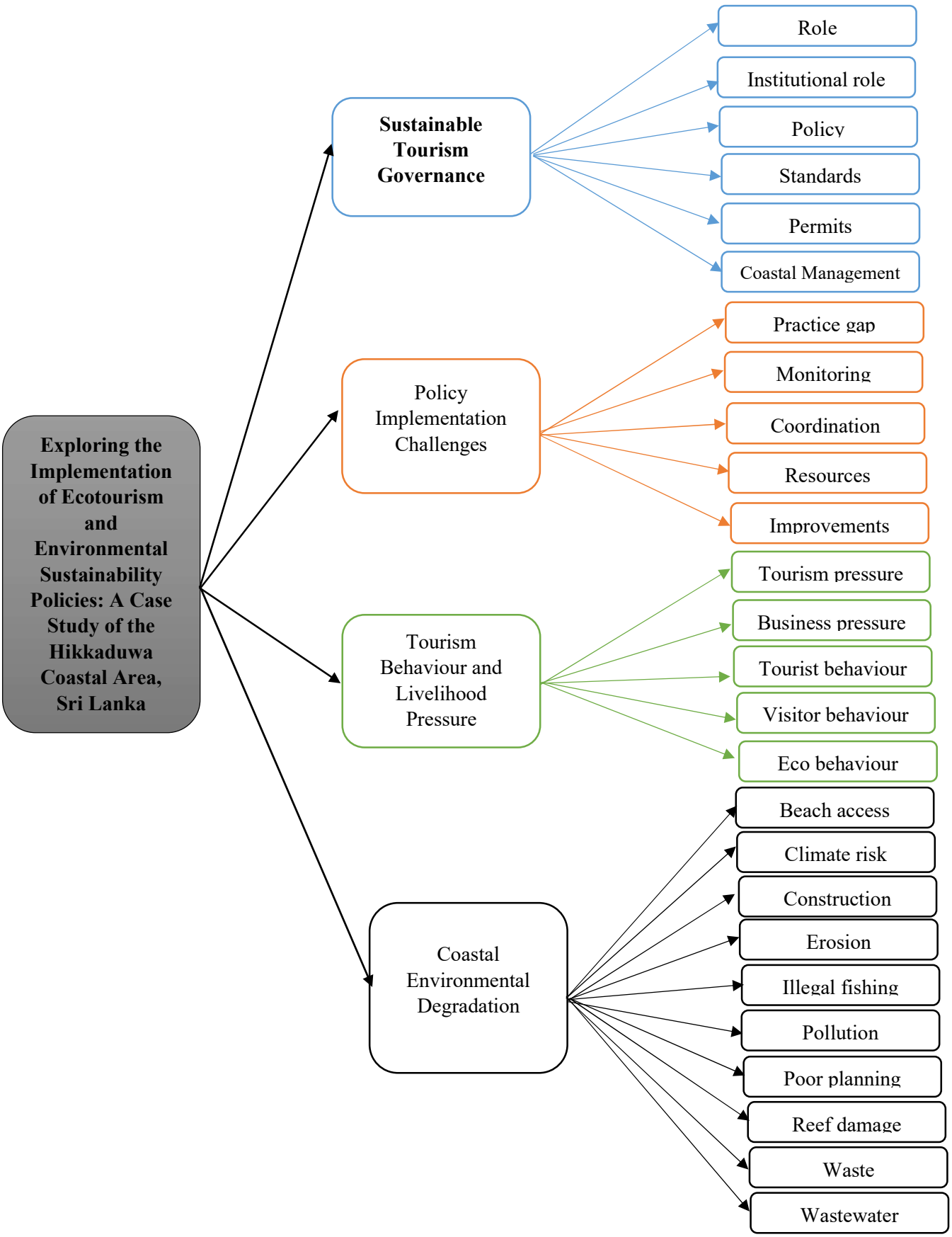



*Figure 1 Thematic matrix from document analysis and interview data*

The thematic matrix contained the themes derived from document analysis and interview data which are presented in Figure 1. The matrix shows four main themes: sustainable tourism governance, policy implementation challenges, tourism behaviour and livelihood pressure, and coastal environmental degradation. These themes were constructed through a process of merging related codes from the documents and interviews. The point is that the matrix is the analytical outline of the findings chapter and helps connect the evidence and the research questions together.

## 5.2 Findings from Policy Document Analysis

This section illustrates the results of documents analysis on official policy, management and audit documents; and involves the use of selected academic studies as supporting evidence. The relevant formal documents were analyzed to determine the existing policies, institutional roles and regulatory mechanisms relating to sustainability and ecotourism in Hikkaduwa. These ranged from coastal management planning, to permit procedures, setback rules, environmental assessment requirements, pollution control and stakeholder coordination in the management of protected areas. The academic studies were utilized in the analysis to reinforce the discussion regarding the policy issues as presented in the local context, such as degradation of the coral reef, coastal erosion, pressure form tourism, tourist behaviour, unplanned development, waste and wastewater issues. Hence this section is objective more than just a description of documents, it actually examines what sustainability issues and management tools are documented in the policy texts and supporting evidence for Hikkaduwa. Most of the findings outlined in this section focus upon Research Question 1.

| No. | Document | Key findings |
|---|---|---|
| 1 | (Coast Conservation and Coastal Resource Management Department, 2025) | Sri Lanka has a formal coastal management framework covering shoreline management, coastal ecosystem conservation, coastal pollution control, Special Management Areas, permit procedures, setback rules and EIA/IEE requirements. The plan also recognises pressure from climate change, habitat degradation, pollution and development demands. |
| 2 | (Jayaratne *et al.*, 2023) | Hikkaduwa Marine National Park shows a declining visitation trend mainly linked to coral bleaching. Coral regeneration is slowed by continuous anthropogenic activities. Visitors are not willing to pay when corals are bleached and broken, but they are willing to pay LKR 322.52 for improved boats and safety jackets. |
| 3 | (Jayarathne, Jayasekara and Gunawardena, 2025) | In Hikkaduwa National Park, pro-environmental behaviour positively influences perceived value. Motivation directly and indirectly affects perceived value. Environmental knowledge alone does not positively influence pro-environmental behaviour or perceived value. This means visitor motivation and behaviour are more important than awareness alone. |
| 4 | (National Audit Office, 2020) | The report identifies unauthorised and informal coastal constructions, wastewater discharge, solid waste disposal, damage to coastal ecosystems, salinity changes and blocked estuaries. It also finds weaknesses in managing adverse impacts and states that stakeholder agencies' operational contribution is not sufficient. |
| 5 | (Nisansala, Jayasinghe and Ranwella, 2021) | The study is useful for showing that anthropogenic activities and unplanned tourism-related development have changed the Hikkaduwa coastal landscape. It supports the finding that local environmental pressure is linked to human activity, coastal infrastructure and weak planning control. |
| 6 | (OCPP, 2025) | The review identifies major threats to Hikkaduwa Marine National Park: coastal infrastructure, habitat destruction and fragmentation, tourism, illegal fishing, climate change, agricultural run-off and pollution. It also states that the park protects one of Sri Lanka's most iconic fringing coral reef ecosystems. |
| 7 | (Silva *et al.*, 2022) | Hikkaduwa has developed rapidly without proper planning, causing coral reef degradation, poor coastal water quality, traffic congestion, resource degradation, poor walkability, blocked sea views and unsuitable built forms. The study recommends design guidelines, better alleyways, parking, sea views, green facades and height controls. |
| 8 | (Weerasekara and Illeperuma, 2024) | From 2001–2023, erosion became dominant across the southwestern coast, with 2,192 out of 3,495 transects showing erosion. Hikkaduwa is identified among the areas experiencing significant erosion. Future predictions for 2034 and 2044 still show erosion concerns around Hikkaduwa, Galle and Kalutara. |

*Table 2 Descriptive summary of documents and supporting sources analysed*

| Code | (Coast Conservation and Coastal Resource Management Department, 2025) | (Jayaratne et al., 2023) | (Jayarathne, Jayasekara and Gunawardena, 2025) | (National Audit Office, 2020) | (Nisansala, Jayasinghe and Ranwella, 2021) | (OCPP, 2025) | (Silva et al., 2022) | (Weerasekara and Illeperuma, 2024) |
|---|---|---|---|---|---|---|---|---|
| Policy | | | | | | | | |
| Coastal management | | | | | | | | |
| Permits | | | | | | | | |
| Institutional role | | | | | | | | |
| Practice gap | | | | | | | | |
| Coordination | | | | | | | | |
| Tourism pressure | | | | | | | | |
| Reef damage | | | | | | | | |
| Visitor behaviour | | | | | | | | |
| Eco behaviour | | | | | | | | |
| Pollution | | | | | | | | |
| Illegal fishing | | | | | | | | |
| Construction | | | | | | | | |
| Poor planning | | | | | | | | |
| Erosion | | | | | | | | |
| Climate risk | | | | | | | | |
| Improvements | | | | | | | | |

*Figure 2 Document analysis code matrix*

The above figure presents an overview of the key documents and supporting sources in the document analysis and the distribution of the main codes in the analysis across the documents and sources. These two sets of tables present a broad policy framework relevant to the management of Hikkaduwa, which includes coastal management, permits, conservation, pollution control and standards for management of protected areas and tourism. They also highlight that similar issues re-emerge from source to source, particularly damage to the reefs, tourism pressure, construction, pollution, erosion and implementation gaps.

### 5.2.1 Sustainable tourism governance in policy documents

The document analysis reveals that the governance of sustainable tourism in Hikkaduwa is not exclusively driven by one policy, but rather by various elements related to the coast and sustainable tourism as well as those of conservation and planning. The Sri Lanka Coastal Zone and Coastal Resource Management Plan 2024-2029 most clearly positions coastal management as a shared national responsibility including shoreline management, conservation of coastal ecosystems, coastal pollution control, Special Management Areas and regulatory controls. The coastal zone, in addition, is utilized for tourism and recreation, livelihood generation, biodiversity, and economic activities and is, however, facing stresses from climate change, the degradation of habitats, pollution and the pressures of development (Coast Conservation and Coastal Resource Management Department, 2025). This indicates that the concept of balancing tourism development with environmental protection is already taken into account in the policy framework.

The documents show evidence of the code of institutional role throughout the document. The Coast Conservation and Coastal Resource Management Department is singled out as the principal organization for coastal zones management and the pioneering institution for coastal conservation under the Coast Conservation and Coastal Resource Management Act under the National Audit Office (2020). The Marine National Park is a protected area, however, this is a concern of Hikkaduwa Committee also as was the Department of Wildlife Conservation. According to the OCPP/DWC review, the Hikkaduwa Marine National Park legally protected since 1979 and has an on-ground DWC office to monitor activities, enforcement and coordination activities with other organizations (OCPP, 2025). This therefore means that governance in Hikkaduwa is not in the hands of a single factor, but institutionalised and shared by actors.

The codes of permits, standards and coastal management are also strongly reflected. The Coastal Zone and Coastal Resource Management Plan outlines the rules and controls for Coastal setbacks and Conservation Areas as well as actions that are prohibited on the foreshore, with rules for Council Permits, EIA/IEE requirements and controls for public access to the foreshore (Coast Conservation and Coastal Resource Management Department, 2025). Research indicates that the built environment, walkability, sea views, coastal water quality, and coral reef condition at Hikkaduwa have suffered from rapid tourism development and lack of planning making these tools relevant (Silva *et al.*, 2022). Standards are also associated with safeguarding the safety of visitors and the quality of tourism. Authors Jayaratne *et al.* (2023) highlight that visitors are willing to pay for better boats and better safety jackets, suggesting that in sustainable governance, not only is there a need for conservation controls, there must also be a service component to reef-based tourism.

It can be stated that the documents point to the existence of formal policy instruments covering Hikkaduwa, such as Coastal Management Planning, Permits, Setback approvals, Conservation Controls, Protected Area management and EIA/IEE protocols as well as tourism protocols. This directly answers the main objective RQ1 which was to arrive at the key policies and governance mechanisms of ecotourism and environmental sustainability relevant to the coastal community of Hikkaduwa.

### 5.2.2 Policy implementation mechanisms and challenges in documents

The document analysis indicates that the key challenge is related not to the lack of implementing mechanisms but in their implementation in a consistent manner in the context of Hikkaduwa. The Coastal Zone and Coastal Resource Management Plan offers various formal procedures such as permit process, setback requirements, coastal pollution control, Special Management Areas, EIA/IEE process, set or ecosystem conservation-related approaches for shoreline management (Coast Conservation and Coastal Resource Management Department, 2025). On the policy level these mechanisms are looking more complete. The comparison of those documents compared to audit results and studies and research undertaken based in Hikkaduwa, however, shows a gap in practice between the expectations on the one hand, and the practice that actually occurs in the coastal area on the other hand.

The NAO report is the best indicator of this disparity. It points out informal and unauthorised constructions, discharge of wastewater, disposal of solid waste, harming coastal ecosystems, changes in salinity and blocked estuaries. More importantly it indicates that there are weaknesses in the management of adverse effects and in implementation of proposed activities. It additionally highlights the fact joint operation of the stakeholder agencies in the field of coastal zone management is not adequate (National Audit Office, 2020). This confirms that lack of implementation is indeed a local behavioral issue, but also an institutional issue of monitoring, coordination and follow-up.

This is corroborated by the OCPP/DWC review of the Hikkaduwa Marine National Park. It lists threats like coastal infrastructure, habitat loss, tourism, illegal fishing, climate change, agriculture run-off/pollution etc. It simultaneously suggests greater enforcement effort, more DWC staff and training, improved regulation of tourism (zoning and carrying capacity controls), new centralised monitoring and new, more inclusive governance with communities and stakeholders (OCPP, 2025). This indicates that protection is acknowledged but the resources and coordination needed for existing protection management systems should be strengthened.

Challenges to implementation also emerge from academic studies. Silva *et al.* (2022) state that the town of Hikkaduwa has developed too rapidly without adequate planning process and proposes to provide design guidelines, change the allocation of some streets, upgrade parking facilities, apply some green façade in the town, control heights and secure sea views. Similarly, Nisansala, Jayasinghe and Ranwella (2021) associate landscape change with tourism related activities of human and unplanned development. Jayarathne, Jayasekara and Gunawardena (2025) demonstrate a need for coral restoration and better visitor facilities to preserve the tourism value of the Marine National Park and they demonstrate that environmental knowledge isn't sufficient unless the mandate to act and pro-ecological attitudes are fostered. Weerasekara and Illeperuma (2024) also demonstrate that erosion vulnerability around Hikkaduwa is a continuing problem and needs to be mitigated and managed continuously.

So the documents make it clear that going beyond formal rules is required for implementation. It needs regular monitoring, agencies to coordinate, practical resources, antiques to be controlled by the locals and more enforcement. This is relevant to RQ1 as it suggests the policies that apply to Hikkaduwa, as well as their implementation mechanisms and recognized deficiencies in the policy and supporting documents.

### 5.2.3 Tourism behaviour, standards and livelihood issues in documents

The documents indicate that the tourism activity and living pressure are the main issues are relating to the problem of sustainability in Hikkaduwa. Hikkaduwa Marine National Park is not only an ecosystem protected area but also a tourist and income generation area. According to Jayarathne, Jayasekara and Gunawardena (2025), the park has a diverse fringing coral reef and generates national and local revenue by charging entrance fees and renting out boats. The same study reveals however, that the rate of coral regeneration has been hindered by uninterrupted human activity around the park, while the rate of visitation has reduced mainly due to coral bleaching. A direct link between reef and tourism income; there is a link between the health of the reef and the value of the tourism experience.

Also identified as a key issue is visitors' behavior. The findings of Jayarathne, Jayasekara and Gunawardena (2025) indicated that pro-environmental behaviour and perceived value are positively related in the Hikkaduwa National Park and that perceiving value is not enough to positively influence pro-environmental behaviour in the same context. It doesn't necessarily mean that just providing tourists with information is enough. Several policies and management tools have to stimulate the responsible attitude, the motivation and the call to action.

The importance of this for Hikkaduwa is that reef based tourism will rely on visitor behaviour in the immediate vicinity of the fragile marine habitat. Additionally, tourism is included in the OCPP/DWC review, as one of the main dangers to Hikkaduwa Marine National Park, along with the other threats such as pollution, illegal fishing, climate change and coastal infrastructure, (OCPP, 2025). Silva *et al.* (2022) state that issues from rapid and unplanned tourism development include congestion, degradation of resources, unsustainable built forms and clogged views. These results indicated that impacts of tourism are not restricted to the reef but extend to the entire coastal town.

Meanwhile, activities in the business and local livelihood cannot be neglected in the documents. Safeguarding the coast is essential to the provision of boat services, hotels, restaurants, and shops and other tourism services. Thus, safety jackets, better boats, the guiding of visitors, design controls, planning standards and sustainable tourism practices deserve importance as they tie the fields of environmental protection together with tourism quality and local income. The finding is related to RQ1 as it considers that there has been a need to discuss the policy frame not only for environmental protection, but also in respect of the behaviour of the visitors, the level of standards within tourism, and dependence on the livelihoods in Hikkaduwa.

### 5.2.4 Coastal environmental degradation in documents

The document analysis shows that degradation of the shore environments is the most pronounced, salient problem that was consistently identified in water sources affecting these shores, accounting for eight out of the eight sources. The major issues include damage to the reef; coastal erosion; unplanned construction; pollution; waste and wastewater; illegal fishing; difficulty accessing beaches; and poor planning and climate-vulnerability. Climate change, degradation of habitat, pollution and development pressure are considered major threats of the coastal zone in the Coastal Zone and Coastal Resource Management Plan. There are also management zones for shoreline protection, ecosystem conservation, coastal pollution control, and Special Management Areas and regulatory processes like permits, setbacks and EIA/IEE procedures provided by the Coast Conservation and Coastal Resource Management Department (2025).

The studies conducted in Hikkaduwa reveal strong damage to the reef. Jayarathne, Jayasekara and Gunawardena (2025) demonstrate reduced visitor numbers associated with coral bleaching, and a drop in willingness to pay by visitors for the Marine National Park when corals are bleached and broken. Tourism and illegal fishing, pollution, coastal infrastructure and habitat fragmentation are identified as threats while the park is one of the iconic fringing coral reef ecosystems in Sri Lanka (OCPP, 2025). It is also supported by Jayarathne, Jayasekara and Gunawardena (2025) who presents that ecotourism sites are vulnerable and pro-environmental behaviour must be practiced to reduce human impact to a certain level.

There's also the recurring theme of unplanned development and bad planning. Silva *et al.* (2022) have indicated that Hikkaduwa has been growing rapidly without planning resulting in problems such as coral reef degradation, bad coastal water quality, traffic congestion, resource degradation, walkability, sea view obstruction and unsuitable built forms. The finding that anthropogenic activities have altered the coastal landscape of Hikkaduwa is also supported by Nisansala, Jayasinghe and Ranwella (2021). All of these sources indicate that not only do tourists visiting the reef area cause environmental problems, but that the physical development of the coastal town causes environmental problems.

Important too are waste, wastewater and pollution. According to National Audit Office (2020), Wastewater discharge, solid waste disposal, damage to coastal ecosystems and blocked estuaries are coastal management concerns identified. Coastal pollution control, encompassing tourist pollution, solid waste and waste water pollution is also addressed in the Coastal Zone and Coastal Resource Management Plan. The problem is complicated further by erosion. According to Weerasekara and Illeperuma (2024), erosion has prevailed over all south-west coast excluding the North central & North western coasts from 2001 to 2023, with Hikkaduwa being one of the areas with significant erosion.

This finding relates to the purpose of the RQ1 question since it highlights the key environmental sustainability problems that are addressed in the policies and management documentation within the related context of Hikkaduwa. The documents indicate there is a series of problems to address in the policy framework, that is, damage, waste, construction, and erosion, not solved separately for the reef.

## 5.3 Findings from Interviews

The results of stakeholder interviews are presented in this section. The interviews helped to gain a deeper understanding of the experiences of various stakeholders in tourism governance and coastal management, as well as the local public services and boat-based tourism in Hikkaduwa, in relation to the implementation of ecotourism and environmental sustainability policies. The interviews revealed the "how" of the implemented policy tools, whereas the document analysis gave insight in the formal policy tools and duties. The results are themed in line with four themes of the thematic matrix; namely sustainable tourism governance, policy implementation challenges, tourism behaviour and livelihood pressure, and coastal environmental degradation. This helps the analysis of the interview results to be compared with the document analysis process. The below table is a summary of the key themes from the full interview coding table, and is included here to aid the discussion. Table below is a condensed version of the key themes, codes and selected evidence presented in this section, and is included here for ease of discussion.

| **Theme** | **Main codes** | **Selected interview evidence** | **Link to RQ** |
|---|---|---|---|
| Sustainable tourism governance | Role, policy, standards, permits, coastal management | P1: "There is also sustainable tourism certification."<br>P2: "Some activities need permits." | RQ2 |
| Policy implementation challenges | Practice gap, monitoring, coordination, resources, improvements | P2: "Rules are already there, but applying rules in daily life is the hard part."<br>P2: "Monitoring is also difficult. We cannot be everywhere every day."<br>P3: "Local authority has many duties, but money is limited." | RQ2, RQ3, RQ4 |
| Tourism behaviour and livelihood pressure | Tourism pressure, business pressure, tourist behaviour, visitor behaviour, eco behaviour | P1: "Hikkaduwa has many small problems that become big because many people use the same area."<br>P4: "This boat work is our income. My family also depends on this." | RQ4 |

| | | | |
|---|---|---|---|
| Coastal environmental degradation | Waste, wastewater, reef damage, construction, beach access, pollution | P4: "Now some coral areas look broken or white colour."<br>P3: "Sometimes drains get blocked, and dirty water becomes a problem, especially after rain." | RQ3, RQ4 |

*Table 3 Summary of interview themes, codes and selected evidence*

### 5.3.1 Stakeholder experiences of sustainable tourism governance

From the interview results, it was noted that stakeholders knew about formal policy tools, with knowledge of governance influenced by their institutional and/or practical role. Participant 1 explained the shift from tourist-number-based planning to sustainability-focused tourism governance: *"Before, many people mainly talked about tourist numbers. Like how many tourists came, how much income came. But now we also talk about clean beach, less plastic, good service, safety, and protecting nature. There is also sustainable tourism certification. That is mainly for hotels and tourism businesses to follow good practices. Like saving water, saving electricity, reducing plastic, keeping the place clean, and treating staff and local people in a good way."* This shows that sustainable tourism governance in Hikkaduwa is not only about increasing arrivals, but also about improving environmental care, service quality and business responsibility. However, Participant 1 also explained that applying one standard is difficult because Hikkaduwa includes many small guesthouses, restaurants, shops, boat operators, tuk-tuk drivers and beach vendors.

The forms of sustainable tourism certification, permits and coastal approval were primarily reflected in the policy code. P1 replied, *"There is also sustainable tourism certification" and "Now we talk more about sustainable tourism."* It indicates that tourism authorities officially acknowledge sustainable tourism, namely in the form of certification and standards. The text also however indicates that sustainability is still a maturing process rather than it is already established in all tourism activities.

The standards code was clearly seen in the explanation P1 stated: *"We also help with registration and standards."* P1 also shared that this comes to the main benefit for "mainly for hotels and tourism businesses to follow good practices" to *"saving water, saving electricity, reducing plastic, keeping the place clean."* This reveals that the standards of interests are primarily geared to business practices mainly in hotel, guest houses and tourism establishments. The interviews also demonstrate however that standards do not just relate to accommodation. P4 remarked "Life jackets are also important. Some tourists do not like to wear, but we tell them it is for safety."

The permit and coastal management codes were mainly reflected in P2's interview. P2 explained "*There are rules. If a person wants to build near the beach or inside the coastal zone, they normally need approval."* In addition, P2 added, *"Some activities need permits."* These comments reflect stakeholder awareness of regulation of coastal development. All activities are not, however, effectively controlled due to the existence of permits. This finding corresponds to the RQ2 which reveals the perception of stakeholders regarding the real presence of sustainability policies, standards, permits and coastal rules in Hikkaduwa. It also reveals dividing up of governance around tourism standards and restrictions on the coast.

### 5.3.2 Policy implementation challenges experienced on the ground

The clear finding from the interviews is that there is a clear gap between formal level policy tools and day-to-day implementation, which is recognised by stakeholders. This was most directly reflected from the viewpoint of P2 where they said *"Rules are already there, but applying rules in daily life is the hard part."* This is a significant quote as it demonstrates that the need in Hikkaduwa is not just simply the lack of rules. On the contrary, the challenge is that these rules are not applied uniformly by tourism operators, the construction of facilities along shorelines, behaviour of visitors, waste management and reef based activities. This reinforces the broader finding that the policy framework is present but the extent to which it is effective relies on its grounded implementation.

This practice gap was also observed with respect to coastal constructions. P2 said, “But in practice, some people first build and then try to get approval later.” This example illustrates how formal permit systems might be undermined if it becomes a requirement to be obtained once development has already occurred. This is a real issue in coastal zones like Hikkaduwa where once buildings, extensions or temporary buildings are created near the beach, it becomes difficult to enforce the ruling. It will also put pressure on officials as it may force them to remove or correct the scenario which might adversely impact the local businesses or livelihoods.

Monitoring was also a key challenge. P2 responded, *“Monitoring is also difficult. We cannot be everywhere every day.”* This demonstrates that rules are not only effective if they are enforced, but so too is their capacity to do so. Hikkaduwa becomes a very busy coastal area with hotels, restaurants, vendors, boat operators, and tourists sharing the same space, so is the Hikkaduwa locals. So, monitoring is challenging and particularly difficult on weekends, holidays and during the tourist season. Participant 2 explained the practical difficulty of monitoring in a busy and already developed coastal destination: *“Monitoring is also difficult. We cannot be everywhere every day. Sometimes a small construction happens quickly. Maybe during holiday time or weekend, they put a temporary structure. Later it becomes difficult to control. Also, coordination takes time. Some matters need support from local authority or police or other offices. If everyone does not act at same time, action becomes slow.”* This shows that the implementation problem is not only the absence of rules, but the limited ability to monitor coastal activities regularly and respond quickly when several institutions are involved.

There were clear indications of coordination problems also. P2 said, *“Some matters need support from local authority or police or other offices”* and *“If everyone does not act at same time, action becomes slow.”* These statements reveal that institutional issues are not always distinguishable regarding environmental and tourism issues. Examples are the Coast Conservation Department being responsible for coastal construction, local authority for waste, SLTDA for the standards of tourism, and the police for public order or illegal activity.

Participant 3 described this problem from the local government perspective: *"Unclear responsibility is also a problem sometimes. People complain to us for everything. But some issues are not fully under our office. For example, beach construction, marine park matters, coral reef damage, these are not only local government matters. But public people do not know which office is responsible. They just come to us. So, we have to coordinate, and that takes time."* This shows that local government often becomes the first contact point for public complaints, even when the issue legally belongs to another agency. Therefore, coordination problems are experienced as delays, unclear responsibility and pressure on local officials.

This was particularly apparent at the local government level due to resource constraints. P3 said *"Local authority has many duties, but money is limited." P3 also said, "Workers are also limited"* and *"If one vehicle breaks down, it affects the whole collection plan."* These quotes illustrate that there needs to be the basic operational capability of staff, vehicles and funding for implementation. Sustainability is, thus, not just a policy issue, but a service delivery issue. This finding relates to RQ2, RQ3 and RQ4, as it demonstrates that the difficulty with monitoring, weak monitoring, limited resources, delayed coordination and inconsistency in enforcing rules on the ground, impact the implementation of the policies.

### 5.3.3 Tourism behaviour and livelihood pressure

The interviews demonstrate that the behaviour of tourists and livelihood pressure are a significant reason for the implementation of environmental sustainability in Hikkaduwa. Many activities are concentrated in a small coastal belt, which was termed as a problem related to tourism. P1 said, *"Hikkaduwa has many small problems that become big because many people use the same area."* This quote reveals that it's not just a big problem of the environment, its many little humors. When they occur concurrently in a constrained and limited coastal area, waste, reef contact, beach crowding, boat congestion and temporary stalls and businesses give rise to problems of increasing magnitude.

P3 also linked tourism with season waste issues, saying; *"During weekends, holidays, and season time, waste becomes more."* This demonstrates a change in environmental management in Hikkaduwa, based on visitor numbers. Peak demand means more loads on the system and more resources for local services to manage when the use of these resources is especially high. P4 said, *"In sea based tourism some times it is too many boats waiting and it is too crowded in the sea area".* This reveals that the reef and marine space are also under pressure from tourism, not just when on land.

Tourist activities were determined as a direct threat to the reef, environment. P1 said *"Some people touch it. Some stand near shallow areas."* P1 also responded, "*Some feed fish because they think it is nice."* P4 explained the same issue from daily experience: *"Some tourists want to put hand into water. Some want to touch coral."* P4 continued, *"Some people think coral is like stone."* Poor visitor behaviour is not necessarily associated with a lack of understanding regarding the reef but has a relationship to the manner in which the reef is experienced as an attraction. If visitors perceive coral as something that they can touch and stand near and interact with, the rules of conservation are more difficult.

The pressure of business upon implementation comes into play. P1 said: *"Business people want to earn. That is normal."* P2 continued, saying, *"Tourism is income for many families." "Small vendors also say this is their income."* added P3. The results reveal that local actors were in favor of environmental protection, but they also rely on tourism revenues. Participant 4 explained how competition creates livelihood pressure in daily boat operations: *"Another problem is competition. There are many boat operators. If one person says, "No, today sea is not good," another person may take the tourists. Then we lose money. So, people sometimes take risk. This is real problem. If there is one common rule for all, then better. Like if sea is bad, nobody should go. Then it is fair."* This gives a clearer example of the livelihood conflict because boat operators may understand safety and environmental rules, but competition makes it difficult to refuse trips unless the same rule is applied to all operators. The reef is a source of revenue for boat operators, while protection of the reef may necessitate restrictions on the number of boats operating and restrictions on visitors. This finding is about policy effectiveness, which is part of RQ4, as the results revealed that tourist behavior, business pressures, tourism-dependent local economies, and localized coastlines affect policy effectiveness.

### 5.3.4 Coastal environmental degradation experienced locally

The results of the interviews demonstrate that coastal environmental degradation is perceived as a reality and perceived as a daily phenomenon in Hikkaduwa. One of the concerns that was expressed often was waste. *"So plastic bottles, food packets, cups and other waste can come to beach."* said P2. An example is provided locally by P3: *"You can see plastic bottles, lunch sheets, food packets, cups, coconut shells, and other things near beach or roadside."* However, P4 said, *"Plastic bottles, shopping bags, food packets, those things sometimes come near sea."* One could argue these comments make it clear that waste is not just an environmental abstract and that this is not made to be taken lightly. It can be seen directly on the beach, on roadsides and close to the sea, particularly areas with a high tourist presence, eating and use by the public.

There was also the large problem of wastewater uncovered as a coastal management problem. As P2 briefed, *"If wastewater management is not proper, it can go towards the beach or sea."* P2 also said, *"Dirty water can affect the coast."* The following quotes demonstrate how wastewater management can have a direct impact on coastal and marine quality. P3 confirmed this as issue from the local service and said, *"Wastewater is also an issue."* P3 went on to explain, *"Sometimes drains get blocked, and dirty water becomes a problem, especially after rain."* This demonstrates an interplay of drainage, rain and wastewater issues at local level. In addition, it indicates coastal pollution is, not just uniquely associated with tourist behaviour, but also associated with regular infrastructure issues.

Another significant problem was damage to the reef. P2 commented, *"Coral damage is also important."* Participant 4 described the visible decline in reef condition through long-term local experience: "*When I was small, the coral was more colourful. Even when I started work also, some places were better than now. Now some coral areas look broken or white colour. Some places are like dead coral. Tourists also ask sometimes, 'Is this coral alive?' Then we feel sad because before it was nicer."* This quote provides stronger local evidence of reef degradation. It also shows that environmental decline affects tourist experience because visitors themselves notice and question the condition of the reef. The following quotations illustrate that local tourism operators that regularly work near the reef can see degradation. Important, since it is an ecological resource and the foundation of the boat-based tourism in Hikkaduwa.

Other concerns included construction and beach access. P2 said, *"Some buildings are close to the beach"* and *"Some places have extended little by little." "Some areas have many shops, restaurants, beach chairs and temporary things.",* P3 said. These activities can result in the loss of open beach area and make beach management more challenging. Access for the public also was elevated. P2 also said, *"Beach access is also a problem in some places"* and *"The beach should not be only for hotels or restaurants."* In the same way, P3 said, *"Public beach access can become difficult in some places."*

This finding is related to RQ3 and RQ4 in so far as it demonstrates the link between environmental degradation and lack of control, poor infrastructure, and pressure of the tourism industry and competing uses of the beach space.

### 5.3.5 Differences in Stakeholder Experiences

SLTDA Officer

The SLTDA officer's perspective on policy implementation focused largely on the formal management side of tourism through the lens of standards, certification, registration and sustainable tourism planning. In this context, the key issues were whether or not tourists businesses keep up with the expected sustainable behavior and whether policies are effective enough to drive the industry. The higher level institutional perspective is where sustainable links are made to policy frameworks, destination quality and development and sustainability of tourism.

Coast Conservation Officer

Coastal regulation, permitting, setback requirements, monitoring, and development control were the main avenues for implementation for the Coast Conservation officer. This less design-centred view was more concerned with the preservation of the coastal area against misguided development patterns, erosion, pollution and unwanted construction. Its existence and the inability to enforce it against a population consistently makes enforcement a big problem, especially in a busy tourist hotspot like Hikkaduwa.

Local Government Representative

The local government representative perceived the implementation of the policy as a service delivery problem. The primary difficulties related to waste collection, drainage, public cleanliness, limited staff and limited finance due to the criticising voice from local people. This highlights that local authorities have a significant amount of responsibility on a day to day basis however they may not necessarily possess the capacity to clearly address the impacts of tourism on the environment.

Local Boat Operator

The local boat operator was more directly tied to the livelihood pressure, demand of the tourists, and survival on a day to day basis. So, from here protecting the environment is significant but hard to balance with one's income.

However, sustainability policies are not always consistent with the actual implementation of policies due to factors like tourists' behavior, competition between operators, and damage to the reef and lack of awareness. In this, the expectation of conservation is contrasted with local economic dependence.

## 5.4 Chapter Summary

The results of the policy document analysis and interviews with stakeholders were presented in this chapter. The results were summarised under four main themes: sustainable governance of tourism, policy implementation challenges, tourism behaviour and livelihood pressures and degradation of the coastal environment. Through document analysis, it was found that Hikkaduwa is encompassed by various formal policy and management instruments. They range from coastal management planning, permit systems, setback and distance restrictions, EIA/IEE's, Special Management Areas, protected area management, water pollution control, and tourism standards and suggestions on tourism's sustainability. Other documents pointed to existing institutional and mechanisms functions for handling coastal development, reef protection, visitor activity and environmental impacts that are suitable instruments for doing so.

The results of the interviews however indicated that while such policies and resources exist, they are not necessarily followed up in practice. Stakeholders acknowledged that there are rules, permits, standards and the concepts of sustainable tourism, and pointed out issues in their daily application. Some of these factors were weak monitoring, insufficient staffing, inadequate staff funding, lack of coordination between institutions, lack of action and inconsistent implementation by business and real-world challenges in tourist behaviour control. There were also some issues and concerns that the local populations were facing, including the management of waste and wastewater, the issue of damage to the reef, the lack of access to the beaches and the issue of too many people in coastal areas to construct their homes.

In general, the chapter demonstrates that the sustainability issue in Hikkaduwa is both institutional and practical. There are policies and management mechanisms already in place, but their effectiveness will depend on how well they can be secured in terms of effective implementation, monitoring and cooperation at destination level. Hence, it is concluded that the difficulty in implementing a consistent application of a policy and not the lack of policy is the problem in Hikkaduwa.

# CHAPTER SIX

## 6.1 Introduction

The findings are discussed in relation to the research questions, literature, document analysis and interviews with stakeholders and have been discussed in this chapter. It describes the benefits of the policies of ecotourism and environmental sustainability to Hikkaduwa, how they are implemented and what parameters of governance, stakeholder and practical issues affect effectiveness.

## 6.2 Discussion of Findings in Relation to Research Questions

### 6.2.1 RQ1: Applicable Ecotourism and Environmental Sustainability Policies in Hikkaduwa

The results indicate that both tourism related and coastal environmental policy frameworks cover Hikkaduwa. The analysis of the document revealed various mechanisms that can be used, such as, shoreline management, conservation of coastal ecosystems, control of coastal pollution and Special Management Areas, permit procedures, setback rules and EIA/IEE requirements. These are mainly stated in Coast Conservation and Coastal Resource Management Department (2024) where coastal development and environment protection are officially controlled in Sri Lanka. This applies to Hikkaduwa since the same space along the coast is used by the hotels, restaurants, beach vendors, boat operators, visitors and residents.

The interview results also verified the fact that sustainable tourism governance is currently put into practice. Tourism registration, business standards, sustainable tourism certification, approvals procedures and permits to develop along the beach were referred to. This demonstrates that there are no tools of policy that are missing. Instead, they are distributed among various institutional functions such as tourism regulation, coastal management, local services and marine tourism activity.

This conclusion is correlated with the existing literature, in which the issue of environmental policy and governance has been brought up as the key to sustainability in tourism. In the argument by Becken *et al.* (2020), the policy of tourism needs to be appropriately equated with environmental management to have an impact on practice. The significance of shared governance between tourists, communities and businesses and the government is noted by Roxas, Rivera and Gutierrez (2020). Thus, the response to the Research Question 1 is the fact that Hikkaduwa has relevant policies of ecotourism and sustainability, but these policies are implemented by a number of actors that have to interact.

### 6.2.2 RQ2: Policy Implementation in Practice in Hikkaduwa

The results indicate that the implementation of policies in Hikkaduwa is uneven. There are sustainable tourism certification, tourism standards, coastal permits and approval procedures but, they are not uniformly implemented by all actors. The interview results revealed that there are those tourism businesses that do adhere to sustainable practices and those that do not. Respondents also cited the fact that there are users of the coast who are known to build in advance without prior permission making permits and setback regulations pointless.

This interpretation can be supported by reading the document in question. According to National Audit Office (2020) unauthorised and informal constructions along the coasts, discharge of wastewater, solid waste disposal, and ineffective stakeholder agency contribution are identified. Silva *et al.* (2022) also demonstrate that Hikkaduwa is rapidly developing unplanned, which results in the degradation of coral reefs, low-quality coastal waters, traffic jams and inappropriate forms of constructions. These results indicate that the pressures associated with tourism related and coastal development have not been successfully controlled by the implementation.

It is an indication of the literature about the policy-practice gap. Sharpley (2023) suggested that due to the focus on economic growth, the sustainability of tourism governance is usually poor since environmental capacity is neglected. The disconnect between knowledge and policy action featured in Scuttari *et al.* (2025) as well, with evidence/formal planning not always culminating in effective implementation. The Hikkaduwa case is in favor of the argument. There exist policies but their execution relies on monitoring, enforcements, local capacity and compliance of the stakeholders. Thus, the Research Question 2 answer is that in the everyday tourism practice policies are partially applied with inconsistency.

### 6.2.3 RQ3: Institutional, Governance and Stakeholder-Related Challenges

The results indicate that weak institutional fragmentation, inadequate coordination, limited resources, pressures on the business and lack of responsibility are all contributing factors to policy implementation in Hikkaduwa. Tourism issues concerning the coastal areas in Hikkaduwa are not limited to a single agency. Issues that are interrelated are waste, wastewater, reef damage, beach access, construction, boat operations and visitor behaviour but they may fall under different institutional responsibilities.

In the results of the interview, participants identified that they have to better collaborate between offices. They observed also that not all the problems can be resolved through a single institution. Local implementation is also influenced by the lack of workers, lack of money and practice service like waste collection and maintenance of drainage. This implies that institutions can be aware of the sustainability problem and yet they might lack the capacity to effectively respond to the problem.

This finding is supported in the analysis of documents. OCPP (2025) has listed some of the pressures such as tourism, coastal infrastructure, illegal fishing, climate change, agricultural run-off and pollution. The combination of these demands mutual control. National Audit Office (2020) also explains that the operational input of stakeholder agencies cannot be enough.

This connects very well with literature. Another key argument by Wondirad, Tolkach and King (2020) was that poor stakeholder cooperation is a significant deterrent to sustainable ecotourism in developing nations. Bichler and Lösch (2019) also mentioned that collaborative governance needs trust, leadership and coordination. Thus, the response to Research Question 3 is that the reasons behind prevalent governance challenges in Hikkaduwa are largely due to a lack of cohesiveness in responsibilities, few resources, inadequate practical coordination.

### 6.2.4 RQ4: Factors Influencing the Effectiveness of Policy Implementation

The results display that policy implementation in Hikkaduwa is effective based on a number of practical and behavioural aspects. Frequent observation is not the least as policies cannot be effective unless they are not only printed in some documents and not reviewed at the destination level. The results of the interview revealed that it is challenging to check-in since not all officers can be present daily. This enhances ease of informal building, improper waste management and inappropriate tourism behaviour to persist.

Another significant consideration is the stakeholder cooperation. The document analysis and interviews revealed that hotels, restaurants, vendors, boat operators, tourists and residents were all involved in the sustainability results. In reef-based tourism, visitor behaviour is of special interest. As Jayasekara, Rajapaksa and Gunawardena (2024) demonstrate, pro-environmental behaviour and motivation have an impact on the perceived ecotourism value, whereas environmental knowledge is not inherently associated with behaviour. This implies that tourist education should be grounded on practice and behaviour.

Another area of policy-effectiveness is livelihood dependence. The interview results indicated that tourism has a significant contribution to the income of a number of families. This can be wedded to literature, where Stronza, Hunt and Fitzgerald (2019) stated that conservation versus local development were crucial to the outcomes of ecotourism. Sustainability policies are better placed to succeed in Hikkaduwa because businesses and other local operators are convinced that environmental protection secures future income. Thus, the solution to Research Question 4 is that policy effectiveness can be determined by monitoring, coordination, resources, stakeholder cooperation, visitor behaviour and the possibility to relate the conservation to livelihood.

## 6.3 Overall Discussion: The Policy–Practice Gap in Hikkaduwa

The general result of the research is that the sustainability issue of Hikkaduwa is predominantly policy-practice gap. Document analysis reveals that there are already in place pertinent policies, the coastal management structures, permits, standards and marine protection issues. The outcomes of an interview indicate that these policies cannot be implemented in everyday practice consistently. This gap manifests itself through inconsistent compliance in business, informal construction, lack of monitoring, poor coordination, shortage of resources and on-going environmental erosion.

The literature reviewed is supported by this finding. According to Becken *et al.* (2020), there is usually a lack of effective practical integration in environmental policy in tourism. According to Sharpley (2023), there is also low sustainability when the development of tourism takes precedence over environmental boundaries. This tension is evident in Hikkaduwa whereby destinations rely on beach and reef tourism, yet the same activities are the cause of waste, wastewater, damage to the reef, congestion, beach access issues and coastal space pressure.

It is also revealed in the discussion that implementation should be conceptualized as a local issue of governance. Without institutions aligning their duty and local actors practicing these policies, there can be no effective policies. Thus, the gap in the policy-practice in Hikkaduwa is not just a legal or administrative problem. It is also influenced by the tourism daily behaviour, livelihood requirement and practical ability of the local institutions.

## 6.4 Recommendations

Considerable reinforcement of regular monitoring should be put in place in weekends, holidays and peak tourism seasons with particular attention given to beach construction, waste disposal, activities of boats, coral protection and access to the beaches by people. A local coordination system is needed to be developed on Hikkaduwa that involves the tourism authorities, coast conservation officers, local government, police, marine park related authorities, owners of businesses and boat operators.

Before the reef-based activities, tourist instructions should be enhanced by clear signboards and short verbal directions. Messages are to not touch corals, not feed fish, not use plastic and safety gear. The management of wastes and wastewater ought to be enhanced by the deployment of additional bins, collection of wastewater at the right time, maintenance of drains and intensified regulation of the wastewater discharge to the beach.

## 6.5 Limitations of the Study

There are certain limitations in this study. The main data on the information were gathered among four participants only. Even though they were key stakeholder positions, the research did not involve broader perspectives of hotel owners, tourists, residents, police officers, wildlife officers or informal vendors. The eight chosen documents were also used to analyse the document, so unpublished policy materials or much local authority records are not necessarily represented. The study concentrated solely on Hikkaduwa, thus narrowing down the scope of generalising the results on other Sri Lankan coastal tourism destinations. The study was also qualitative, implying that it also failed to statistically determine the extent of environmental degradation.

## 6.6 Conclusion

The findings were discussed in this chapter, as well as with respect to the research questions, literature, and the document analysis and interviews. The research concludes that the Hikkaduwa has applicable sustainability policies and institutions but these are still not well realized. To decrease the policy-practice gap, there is need to have stronger monitoring, coordination, stakeholder cooperation and realistic local action.

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# Appendix

## Appendix 1 – Interview Questionnaire

### Interview 1: Sri Lanka Tourism Development Authority officer

1. Can you tell me what work you do at the Sri Lanka Tourism Development Authority?
2. Does the Sri Lanka Tourism Development Authority have any plans or programmes for sustainable tourism in coastal places like Hikkaduwa? For example, hotel standards, sustainable tourism certification, or marine tourism plans.
3. In Hikkaduwa, do hotels, guesthouses, restaurants or tour operators follow sustainable tourism practices? For example, reducing plastic, managing waste, protecting the beach, or giving correct information to tourists.
4. What are the main tourism problems in Hikkaduwa now? For example, too many tourists, beach waste, coral reef damage, unplanned businesses, traffic, or poor visitor behaviour.
5. When tourism businesses want to grow, but the environment needs protection, how is this managed?
6. What should be improved to make Hikkaduwa a better sustainable tourism destination?

### Interview 2: Coast Conservation and Coastal Resource Management Department officer

1. Can you tell me what work you do in coastal protection or coastal management?
2. What rules are there for building near the beach in Hikkaduwa? For example, permits, setback distance, approval before building, or environmental checks.
3. In Hikkaduwa, what building or coastal development problems do you see? For example, buildings too close to the beach, illegal structures, blocked beach access, or damage to the coast.
4. Is it easy or difficult to control these problems? What makes it difficult?
5. What are the main environmental problems in the Hikkaduwa coastal area? For example, erosion, waste, wastewater, coral damage, or pollution.
6. What support is needed to protect the Hikkaduwa coast better? For example, more staff, better monitoring, stronger rules, or better coordination with other offices.

**Interview 3: Hikkaduwa local government representative**

1. Can you tell me what work your office does in Hikkaduwa town or coastal area?
2. What are the main environmental problems you see in Hikkaduwa? For example, beach waste, wastewater, traffic, illegal buildings, crowded areas, or poor public beach access.
3. How does your office manage these problems? For example, waste collection, cleaning, inspections, public awareness, or working with other government offices.
4. What makes this work difficult? For example, not enough money, not enough workers, weak public support, business pressure, or unclear responsibility.
5. Do hotels, restaurants, small shops, boat operators and local people cooperate with environmental rules?
6. What should be improved in Hikkaduwa to protect the environment better?

**Interview 4: Local boat**

1. How long have you worked as a boat operator in Hikkaduwa?
2. What changes have you seen in the sea, beach or coral reef over the years? For example, coral damage, less fish, dirty water, more waste, or erosion.
3. Do boat trips affect the reef or sea area in any way? For example, too many boats, anchors, boat movement, tourists touching corals, or waste from visitors.
4. Are there any rules you must follow when taking tourists near the reef? For example, where boats can go, where boats can stop, safety rules, or protecting corals.
5. Are these rules easy or difficult to follow in daily work? What makes it difficult?
6. What help do boat operators need to protect the reef better? For example, training, clear rules, tourist education, fixed boat areas, or support from government offices.

# Appendix 2 – Transcript Coding



## Implementation of Ecotourism

Interview Transcript

Interview 1: Sri Lanka Tourism Development Authority Officer

**Participant:** Participant 1

**Role:** Officer, Sri Lanka Tourism Development Authority

**Interviewer:** Thank you for speaking with me. I will ask few questions about tourism and environment in Hikkaduwa. Can you tell me what work you do at the Sri Lanka Tourism Development Authority?

**Participant:** Yes, okay. I work in tourism related work *[Role]*. Mainly we look at tourism places, tourism businesses, and how they are working. We also help with registration and standards *[Standards]*. Sometimes hotel people or guesthouse people contact us and ask about tourism rules or how to improve their place. My work is not only about Hikkaduwa. We look at many places. But

people also. The coast is the main value of Hikkaduwa. If we lose the beach and reef, tourism will also go down. So, protection must happen now, not after everything is damaged.

Interview 3: Hikkaduwa Local Government Representative

**Participant:** Participant 3

Role: Hikkaduwa local government representative

**Interviewer:** Thank you for speaking with me. I will ask few simple questions about Hikkaduwa town and coastal area. Can you tell me what work your office does in Hikkaduwa town or coastal area?

**Participant:** Yes, okay. Our office mainly looks after local level services in Hikkaduwa area *[Role]*. We deal with things like waste collection, cleaning public places, drainage problems, public roads, small local complaints, and some matters connected with shops and public areas. In a tourism place like Hikkaduwa, this work is not small. Because every day many people use the town. Local people live here, tourists come, hotels operate, restaurants operate, shops operate, and beach area also becomes busy. So, we have to manage both local people's needs and tourism needs. Sometimes people think local government only collects waste, but actually we get many types of complaints. For

Interview 2: Coast Conservation and Coastal Resource Management Department Officer

**Participant:** Participant 2

**Role:** Officer, Coast Conservation and Coastal Resource Management Department

**Interviewer:** Thank you for speaking with me. I will ask some simple questions about coastal protection and

Hikkaduwa area. Can you tell me what work you do in coastal protection or coastal management?

**Participant:** Yes, okay. My work is mainly related to coastal area management *[Role]*. We look at things like coastal development, beach protection, erosion problems, and whether building activities near the coast are done properly. If someone wants to build or change something inside the coastal zone, they need to follow the correct process. So, our department checks those matters. We also look at problems like erosion, damage to beach, sand movement, and some pollution issues also. In places like Hikkaduwa, this work is very important because the coast is used for many things. There are hotels, restaurants, local houses, shops, tourists, boat people, and also the marine park area. So, many people use the same coastal space *[Tourism pressure]*. That is why coastal management is not easy. We

Interview 4: Local Boat Operator

**Participant:** Participant 4

Role: Local boat operator in Hikkaduwa

**Interviewer:** Thank you for speaking with me. I will ask few simple questions about your work and Hikkaduwa sea area. How long have you worked as a boat operator in Hikkaduwa?

**Participant:** I have worked here for about 12 years now. Before that also I was helping my uncle with boat work, but full-time I started around 12 years ago. We do boat rides for tourists, mainly glass-bottom boat rides *[Role]*. Some tourists come to see coral and fish. Some just want to go

little bit to the sea and come back. Most of them are foreign tourists, but local people also come, especially school holidays and weekends. Hikkaduwa is very famous for this, no. Many people come and ask, "Can we see coral?" So this boat work is our income. My family also depends on this *[Business pressure]*. I know many boat people here. Some are doing this for long time, some are new.

**Interviewer:** What changes have you seen in the sea, beach or coral reef over the years? For example, coral damage, less fish, dirty water, more waste, or erosion.